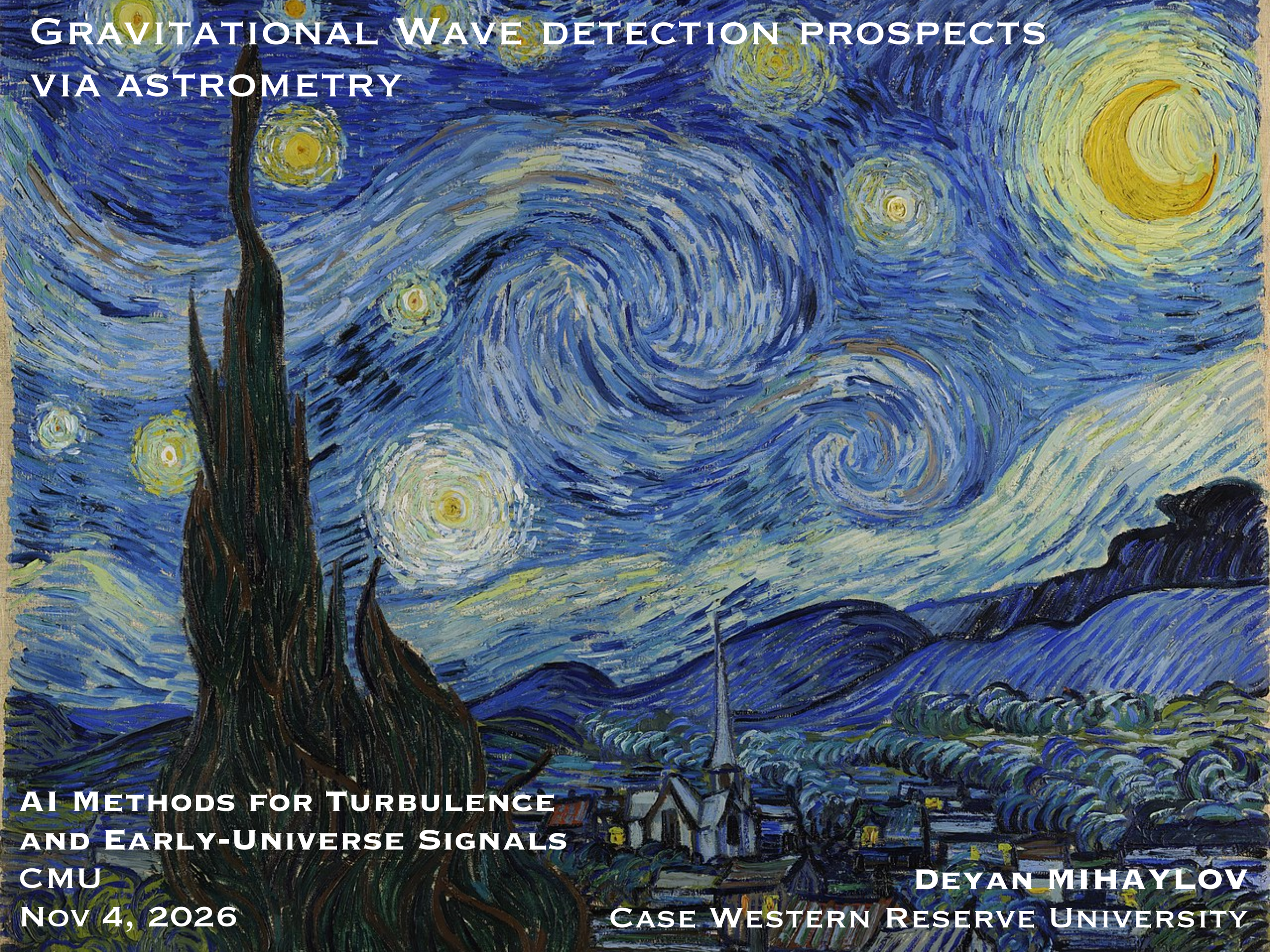
# Gravitational Wave detection prospects via astrometry

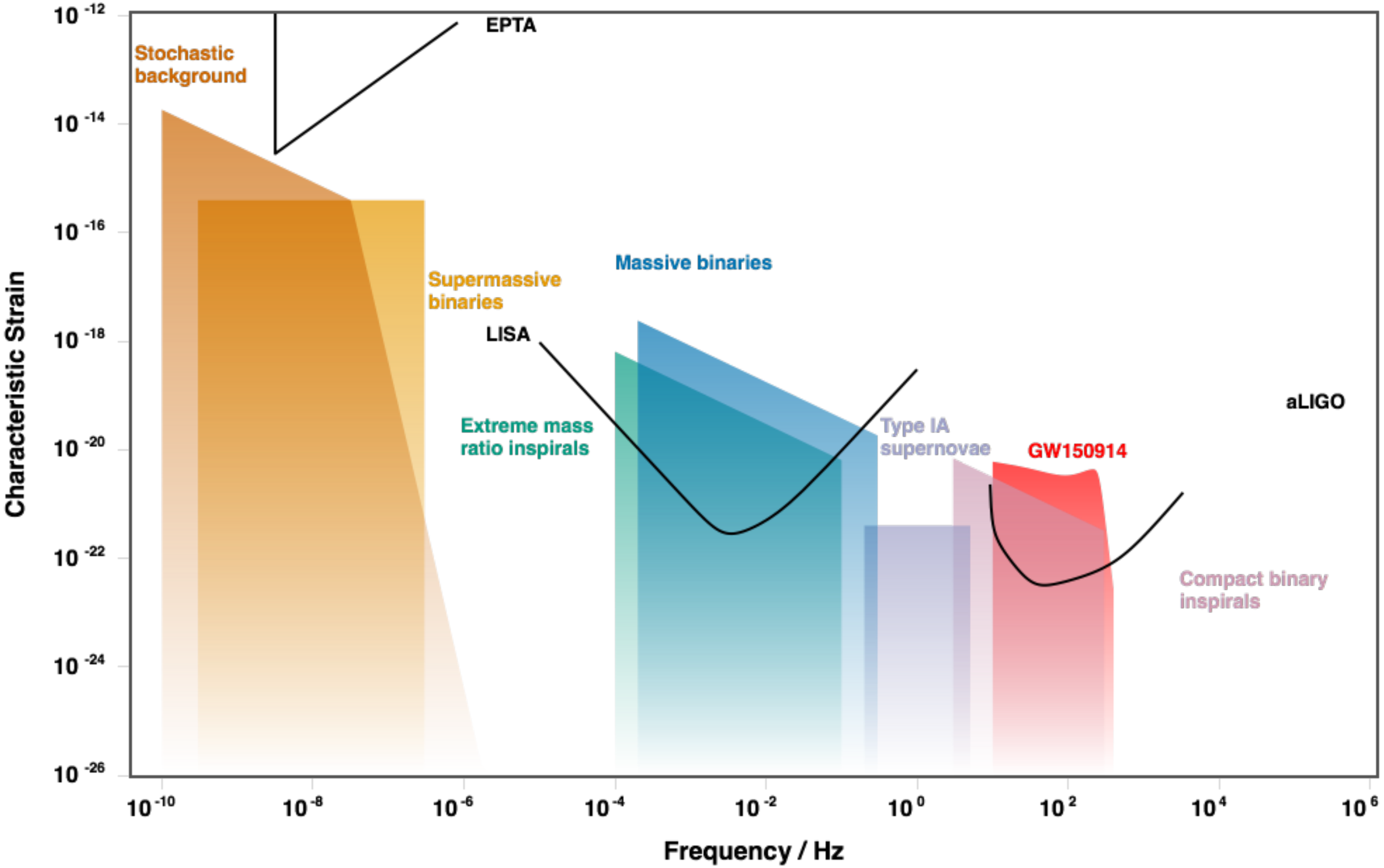


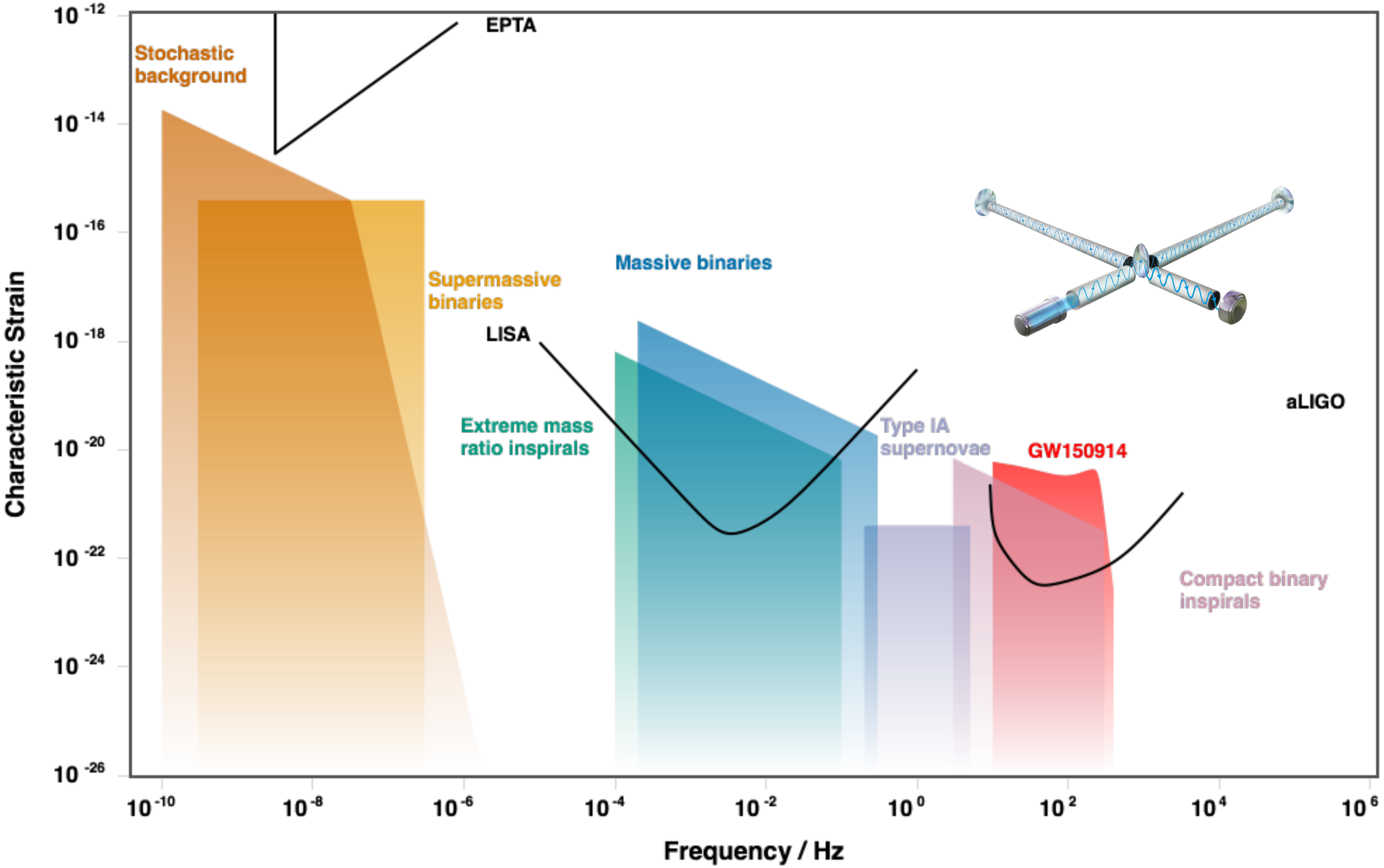
## AI Methods for Turbulence and Early-Universe Signals

CMU

Nov 4, 2026

Deyan Mihaylov
Case Western Reserve University


Characteristic Strain
10 -12
10 -14
10 -16
10 -18
10 -20
10 -22
10 -24
10 -26
Stochastic background
EPTA
Supermassive binaries
Massive binaries
LISA
Extreme mass ratio inspirals
Type IA supernovae
aLIGO
GW150914
Compact binary inspirals
10 -10
10 -8
10 -6
10 -4
10 -2
10 0
10 2
10 4
10 6
Frequency / Hz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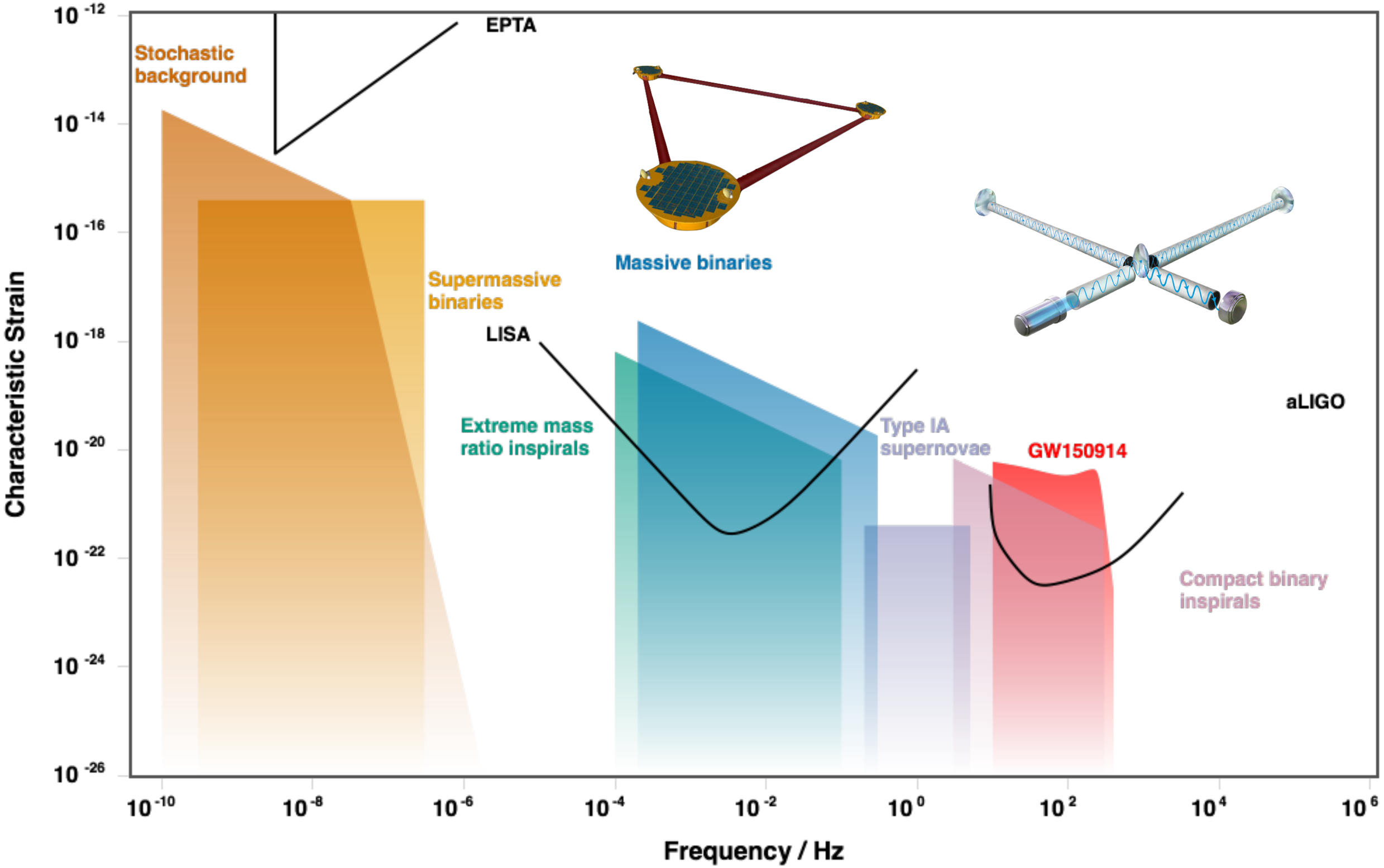
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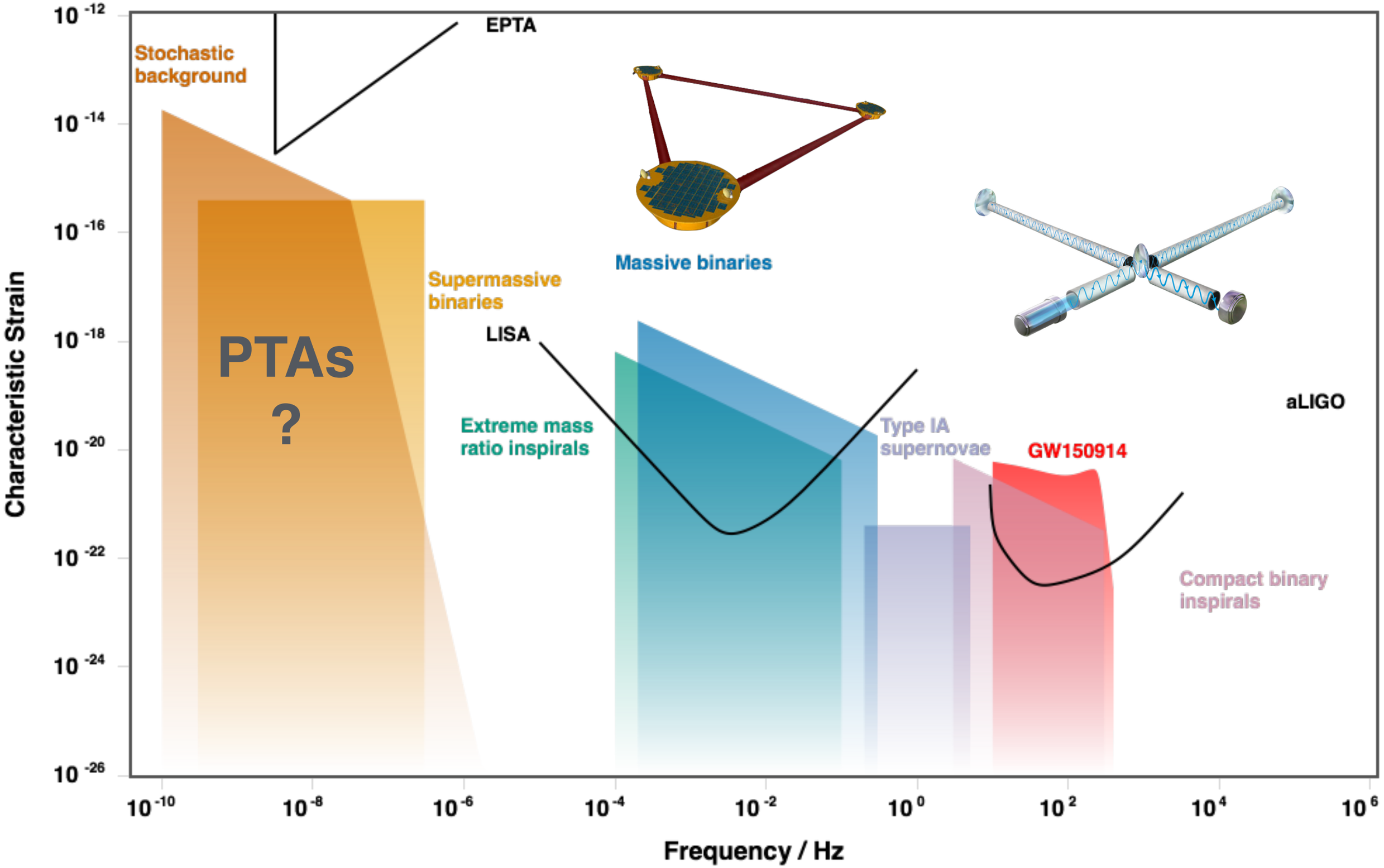
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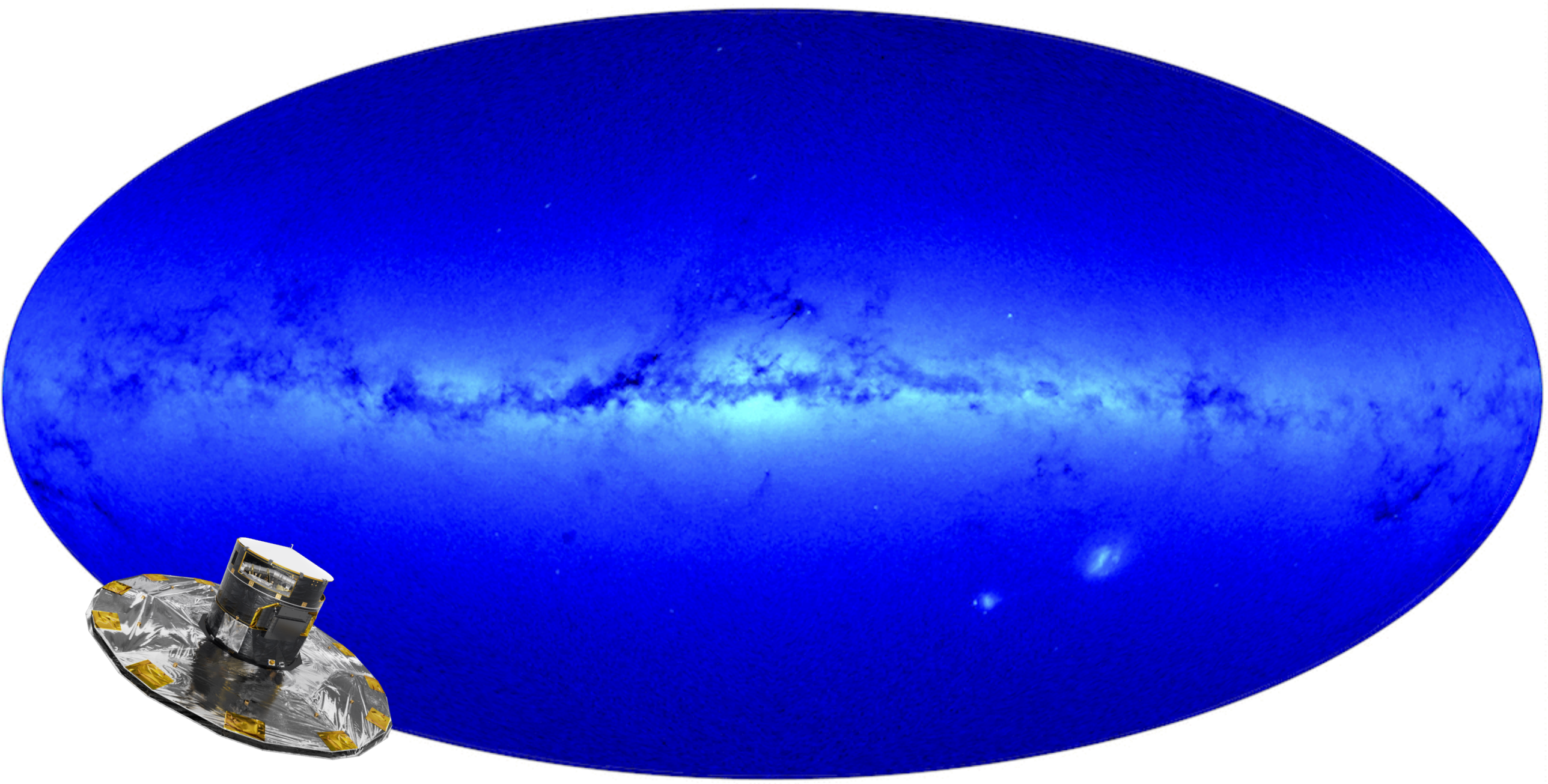


Gerry
GILMORE

Anthony
LASENBY

Jonathan
GAIR

Christopher
MOORE

# Contents

1. Astrometric response of a gravitational wave

2. Sensitivity of Gaia

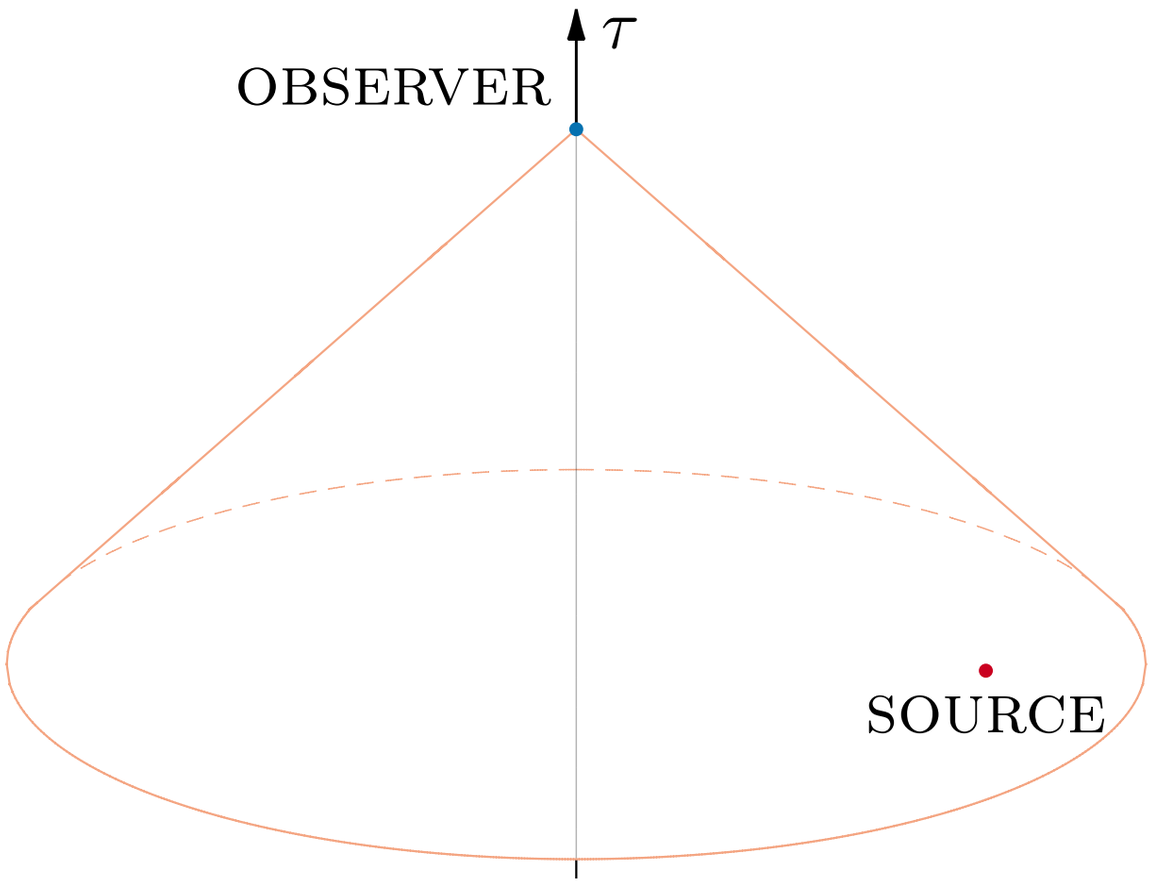
3. Background correlations

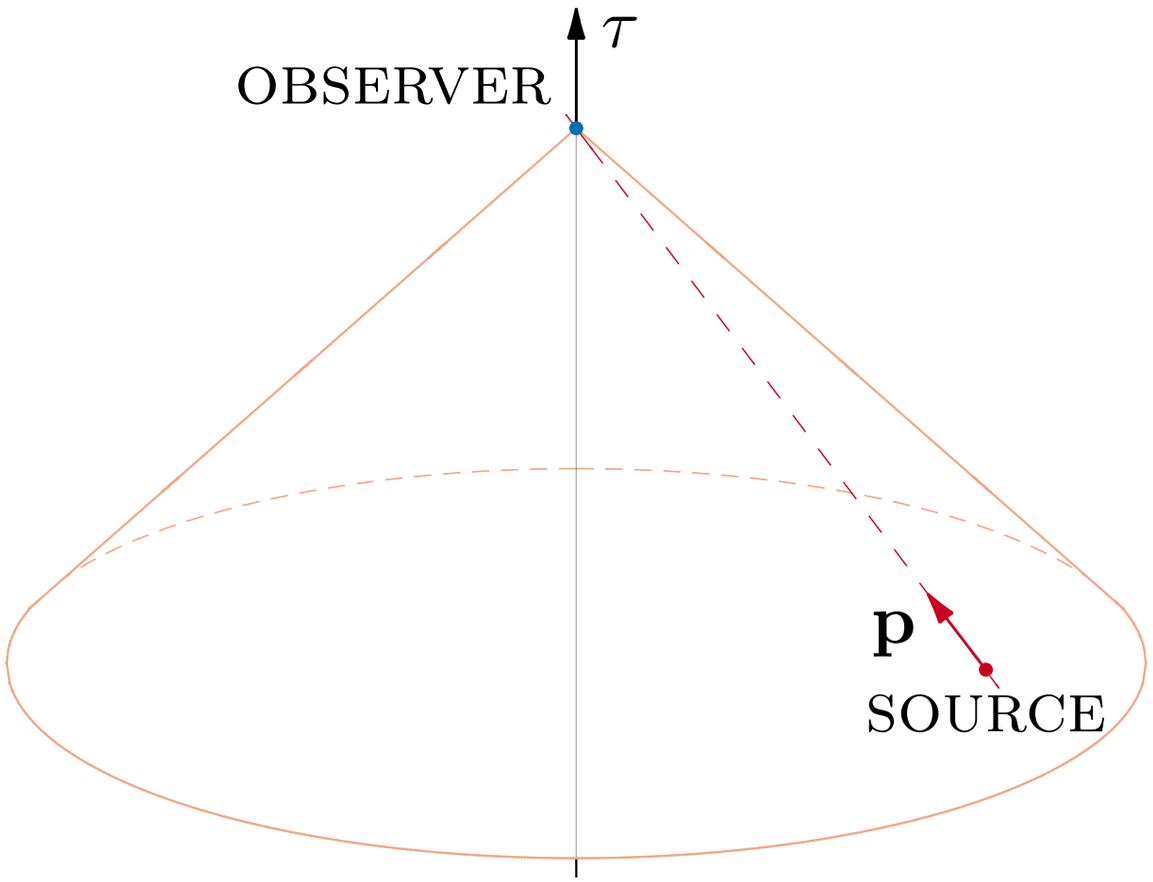
4. Constraining the speed of light

5. New directions

# Astrometric response of a Gravitational Wave

Observer (Earth) and photon source (star) are at rest in flat space

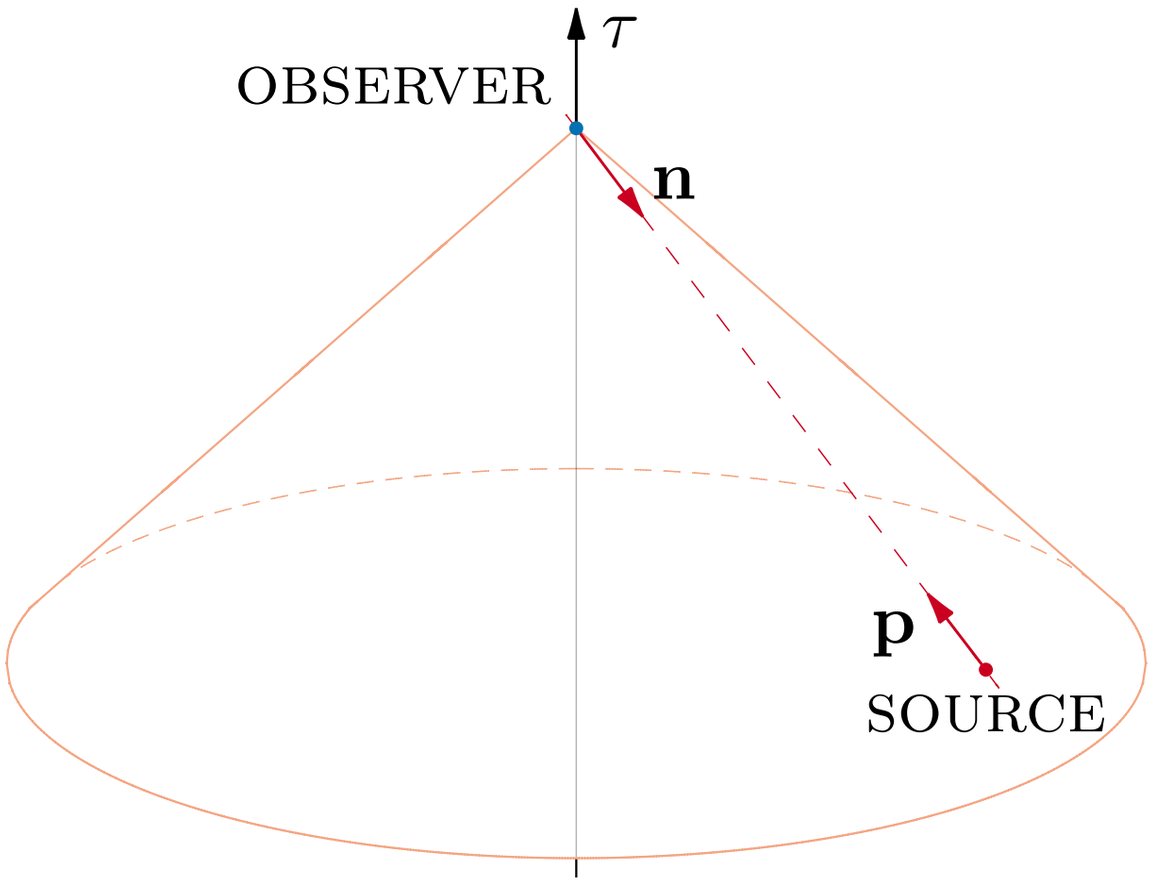





## Observer (Earth) and photon source (star) are at rest in flat space

## Joined by a null geodesic

$$\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2}\, x^\mu(\lambda) = 0, \quad p^\mu = \frac{\mathrm{d}}{\mathrm{d}\lambda}\, x^\mu(\lambda) \equiv \text{const.}$$

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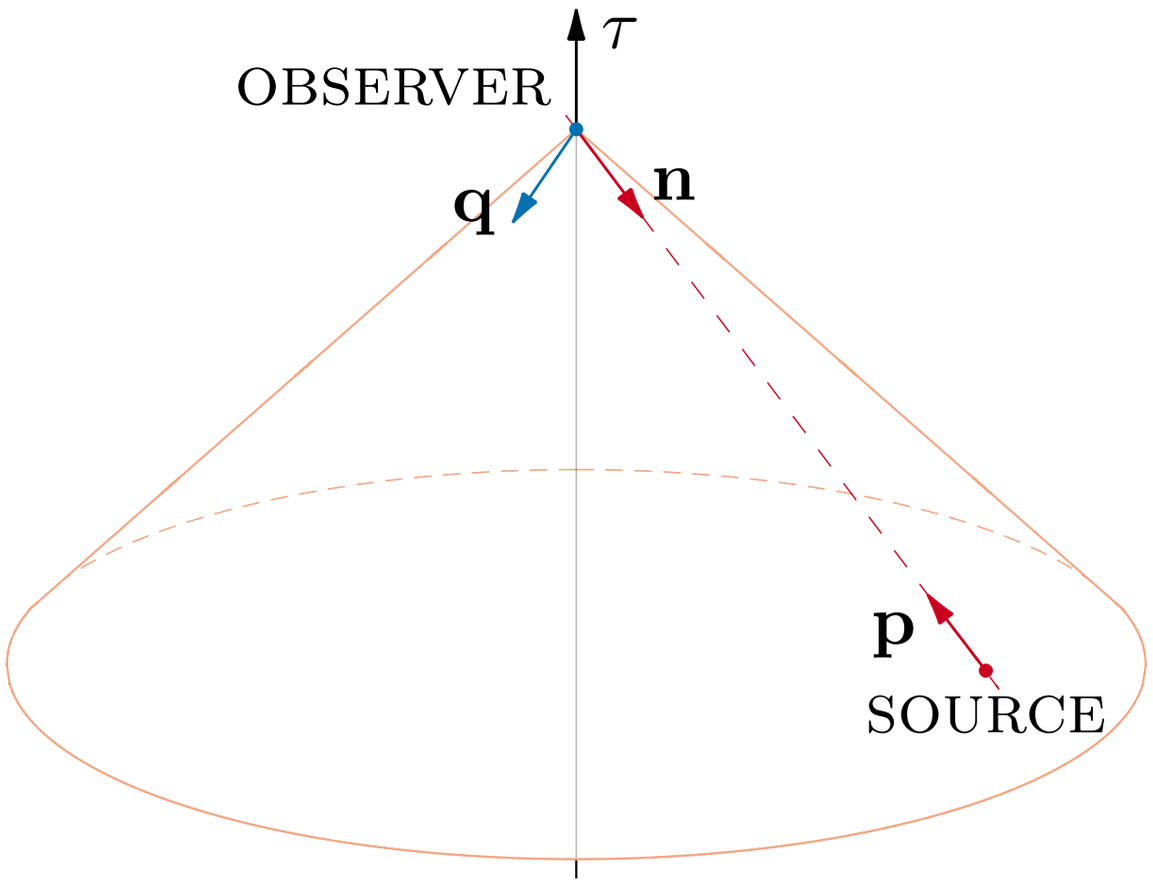
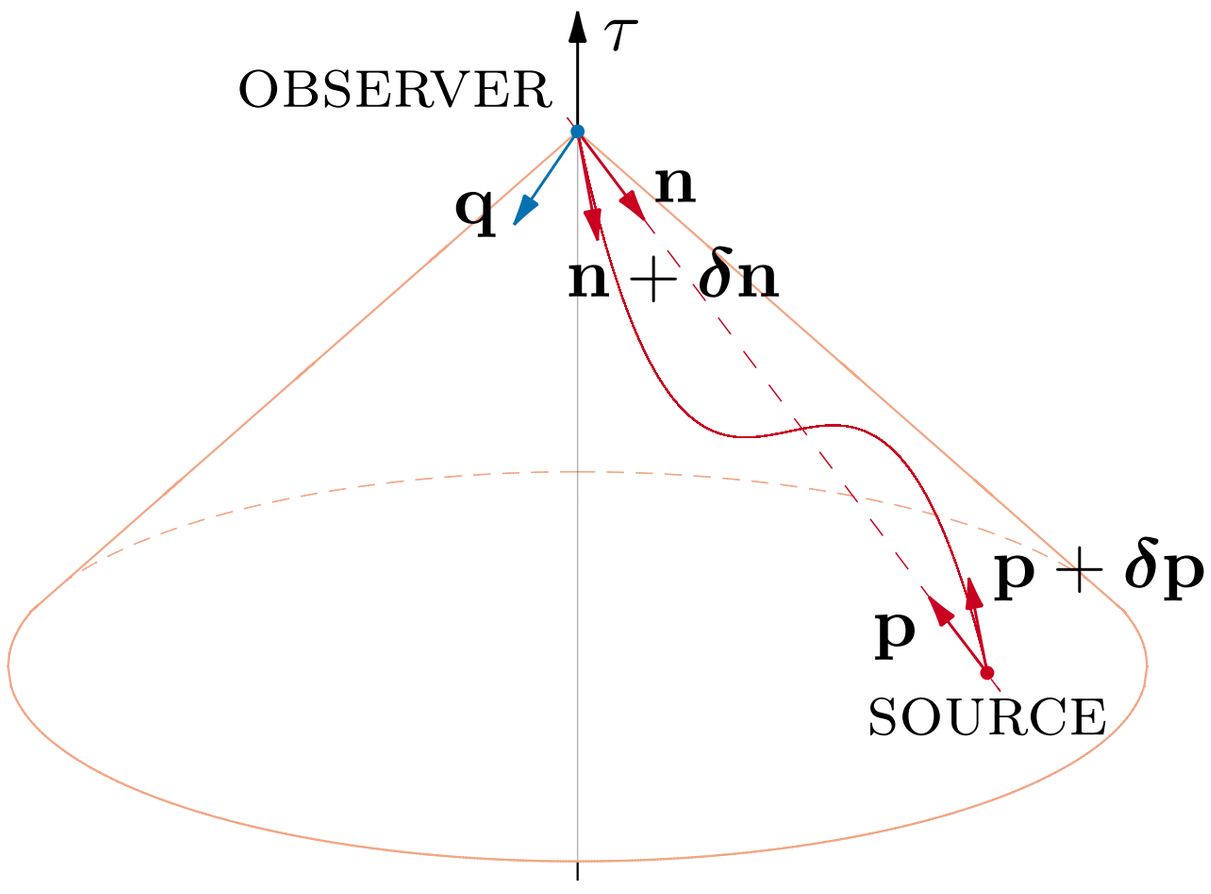
## The Observer measures **astrometric position** and **frequency**

$$n_{\hat{\imath}}, \quad \Omega$$




## Now consider perturbing the flat space-time with a GW

$$h_{\mu\nu}\left(t, x^i\right) = \Re\left\{ H_{\mu\nu} \exp\left(\mathrm{i} k_\rho x^\rho\right) \right\},$$
$$k^\rho = \omega\left(1, -q^i\right)$$

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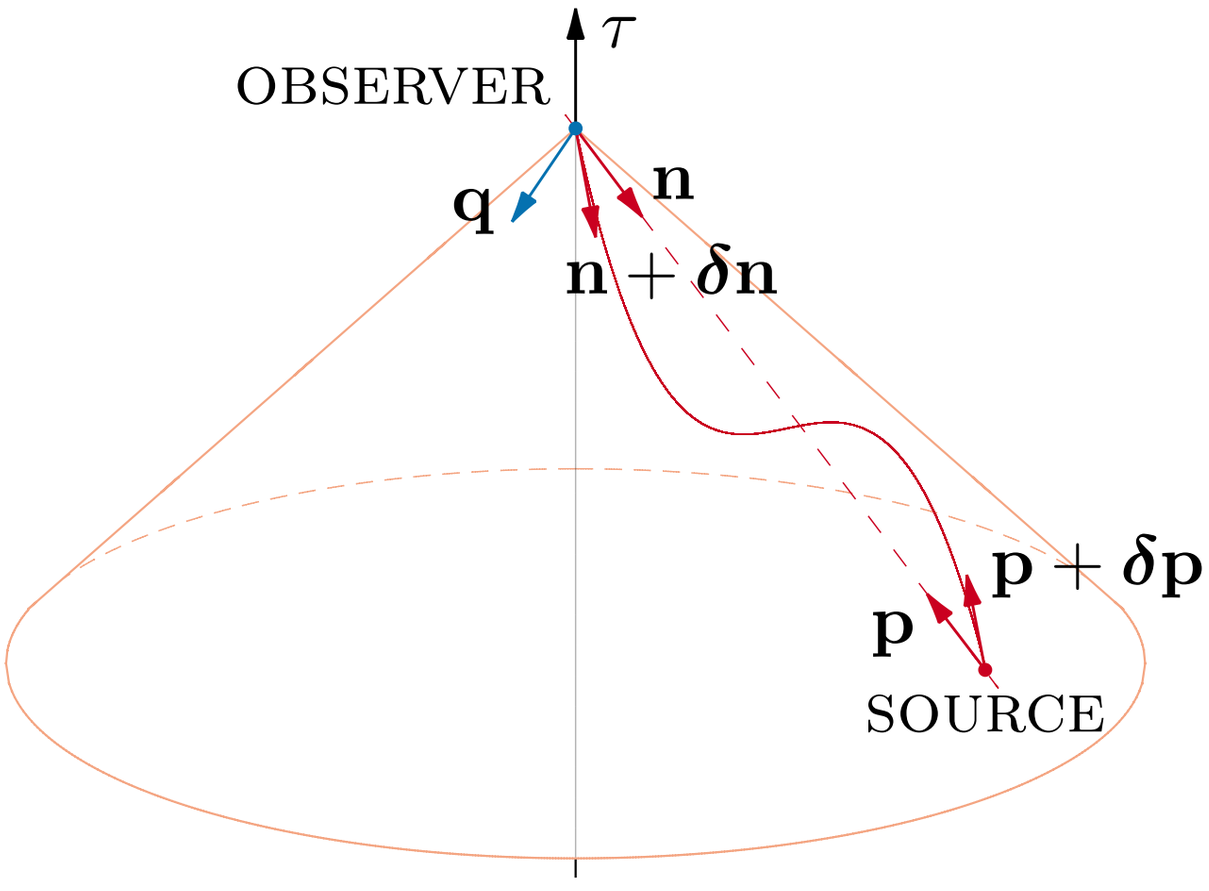
## Worldlines of **observer** and **source** are unaffected

## Photon worldline is a geodesic in both metrics

$$x^\mu(\lambda) \mapsto x^\mu(\lambda) + \delta x^\mu(\lambda)$$

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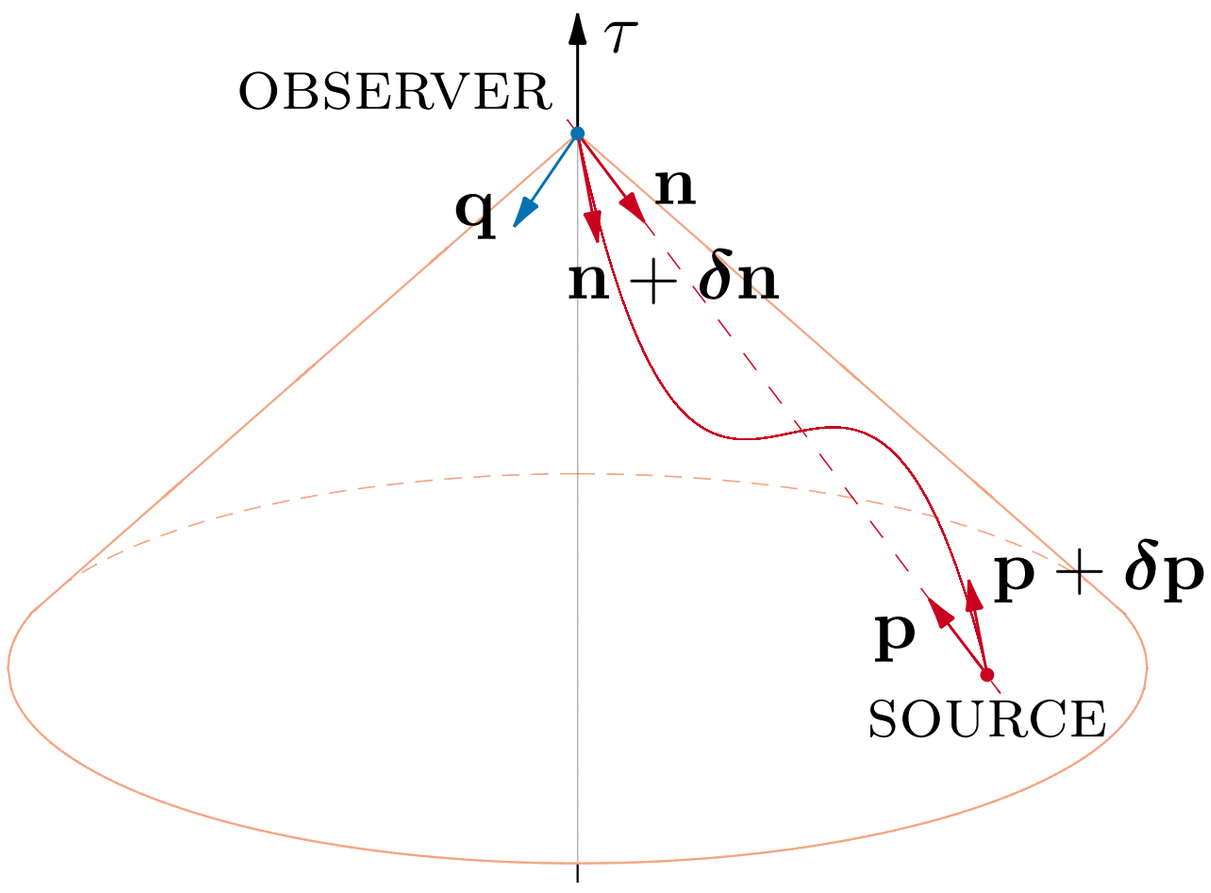
## Evolves according to the parallel transport equation

$$\frac{\mathrm{d}^2}{\mathrm{d}\lambda^2}\, \delta x^\mu_{t_0}(\lambda) = -\Gamma^\mu_{\nu\rho}\, p^\nu p^\rho$$

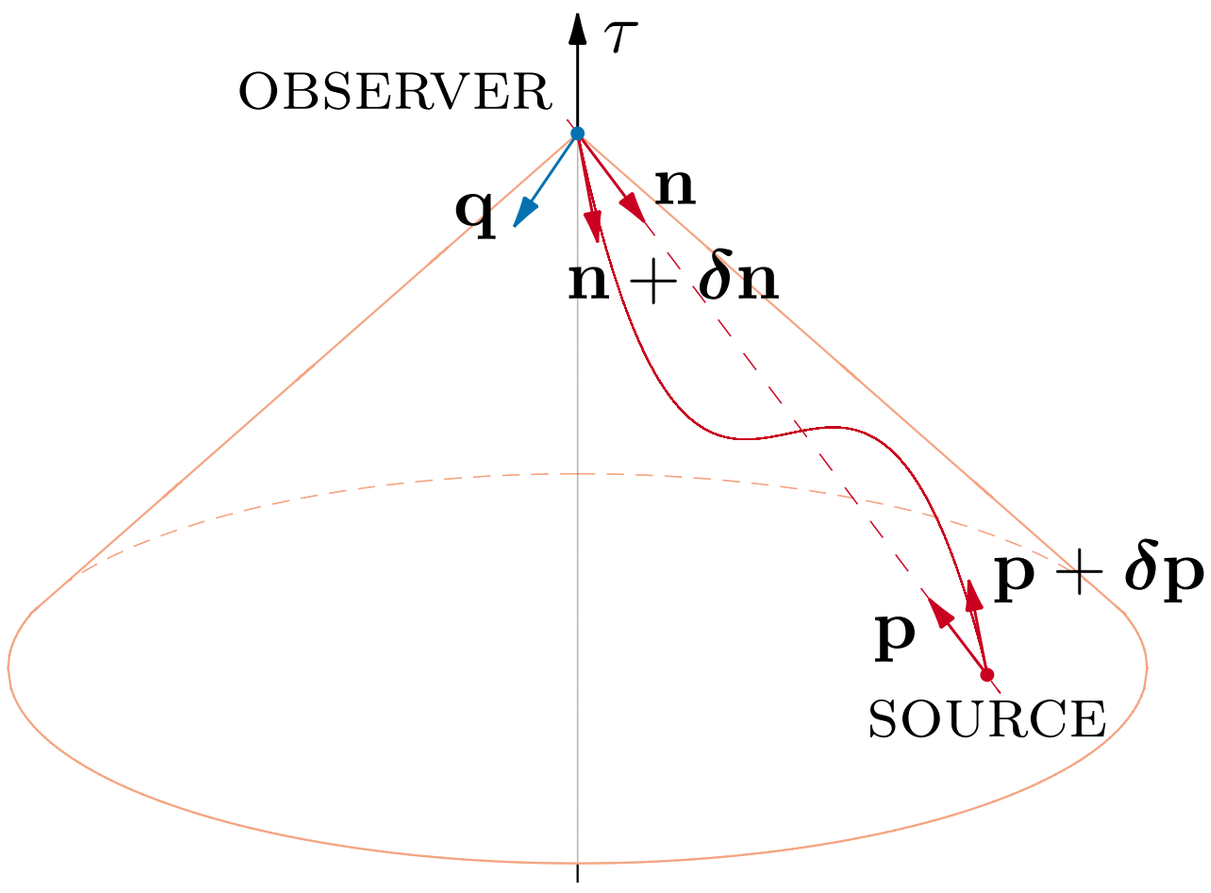



## Integrate along the worldline of the photon

## Boundary conditions:

### A. Photon path is null

### B. Photon path intersects source and observer worldlines




## Integrate along the worldline of the photon

## Boundary conditions:

### A. Photon path is null

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## Perturbed astrometric position and frequency

$n_{\hat{\imath}} + \delta n_{\hat{\imath}}, \quad \Omega_{\rm obs}$

# Redshift

$$z = \frac{n^i n^j}{2\left(1 - n_k q^k\right)} \left[h_{ij}(\text{OBS}) - h_{ij}(\text{SOURCE})\right]$$

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# Astrometric shift

$$\delta n_{\hat{i}} = \Bigg[ \Bigg( \left\{ 1 + \frac{\mathrm{i}(2 - \vec{q}\cdot\vec{n})}{\omega\lambda_{\mathrm{S}}\Omega(1-\vec{q}\cdot\vec{n})} \left[ 1 - \exp\left(-\mathrm{i}\omega\Omega\lambda_{\mathrm{S}}(1-\vec{q}\cdot\vec{n})\right) \right] \right\} n_{\hat{i}} \\ - \left\{ 1 + \frac{\mathrm{i}}{\omega\lambda_{\mathrm{S}}\Omega(1-\vec{q}\cdot\vec{n})} \left[ 1 - \exp\left(-\mathrm{i}\omega\Omega\lambda_{\mathrm{S}}(1-\vec{q}\cdot\vec{n})\right) \right] \right\} q_{\hat{i}} \Bigg) \frac{H_{jk} n^j n^k}{2(1-\vec{q}\cdot\vec{n})} \\ - \left\{ \frac{1}{2} + \frac{\mathrm{i}}{\omega\lambda_{\mathrm{S}}\Omega(1-\vec{q}\cdot\vec{n})} \left[ 1 - \exp\left(-\mathrm{i}\omega\Omega\lambda_{\mathrm{S}}(1-\vec{q}\cdot\vec{n})\right) \right] \right\} H_{\hat{i}j} n^j \Bigg] \exp(-\mathrm{i}\omega t_0) \,.$$

# Astrometric shift

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\begin{aligned}
\delta n_{\hat{i}} = & \Bigg[ \Bigg( \left\{ 1 + \frac{\mathrm{i}(2 - \vec{q}\cdot\vec{n})}{\omega\lambda_{\mathrm{S}}\Omega(1-\vec{q}\cdot\vec{n})} \left[ 1 - \exp\left(-\mathrm{i}\omega\Omega\lambda_{\mathrm{S}}(1-\vec{q}\cdot\vec{n})\right) \right] \right\} n_{\hat{i}} \\
& \qquad - \left\{ 1 + \frac{\mathrm{i}}{\omega\lambda_{\mathrm{S}}\Omega(1-\vec{q}\cdot\vec{n})} \left[ 1 - \exp\left(-\mathrm{i}\omega\Omega\lambda_{\mathrm{S}}(1-\vec{q}\cdot\vec{n})\right) \right] \right\} q_{\hat{i}} \Bigg) \frac{H_{jk} n^j n^k}{2(1-\vec{q}\cdot\vec{n})} \\
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= & \, \delta n_{\hat{i}} \left( h(\mathrm{OBS}), h(\mathrm{SOURCE}) \right)
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$$

## In the distant source limit

$$
\delta n_{\hat{i}} = \frac{n_{\hat{i}} - q_{\hat{i}}}{2(1-\vec{q}\cdot\vec{n})} h_{\hat{j}\hat{k}}(\mathrm{OBS}) n^{\hat{j}} n^{\hat{k}} - \frac{1}{2} h_{\hat{i}\hat{j}}(\mathrm{OBS}) n^{\hat{j}} \,.
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$$

FRACTIONAL ERROR $\delta$

$10^{1}$

$10^{-1}$

$10^{-3}$

$10^{-5}$

$10^{-7}$

ALPHA CENTAURI

CUMULATIVE FRACTION $\eta$

1.00

0.75

0.50

0.25

0

$10^{-2}$ $10^{-1}$ $10^{0}$ $10^{1}$ $10^{2}$ $10^{3}$ $10^{4}$ $10^{5}$

OBSERVER - SOURCE DISTANCE [pc]

# Effect on the sky

# Astrometric shift

$$
\begin{aligned}
\delta n_{\hat{i}} = & \Bigg[ \Bigg( \left\{ 1 + \frac{\mathrm{i}(2 - \vec{q}\cdot\vec{n})}{\omega\lambda_{\mathrm{S}}\Omega(1-\vec{q}\cdot\vec{n})} \left[ 1 - \exp\left(-\mathrm{i}\omega\Omega\lambda_{\mathrm{S}}(1-\vec{q}\cdot\vec{n})\right) \right] \right\} n_{\hat{i}} \\
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\end{aligned}
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= \delta n_{\hat{i}} \left( h(\mathrm{OBS}), h(\mathrm{SOURCE}) \right)
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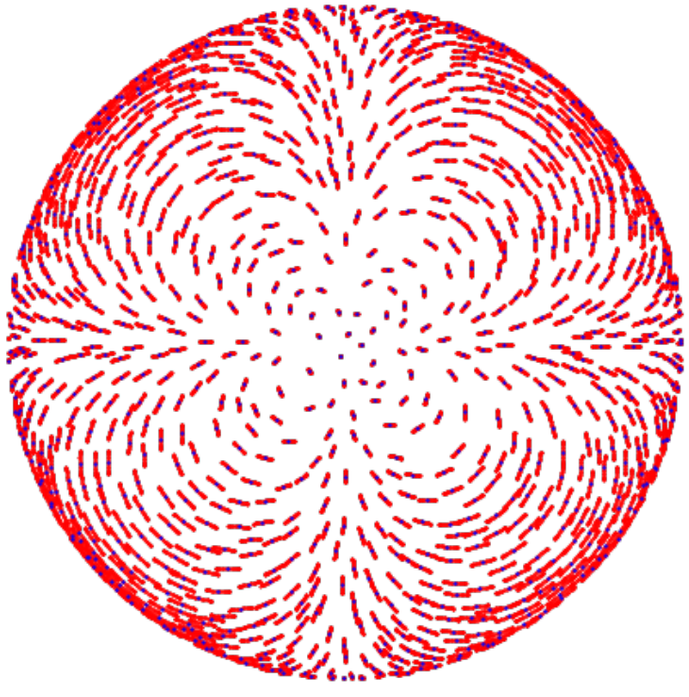
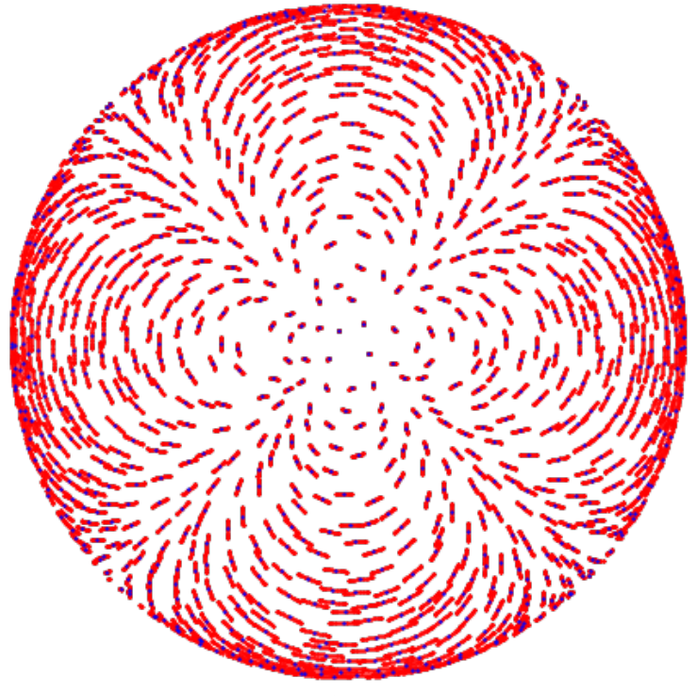
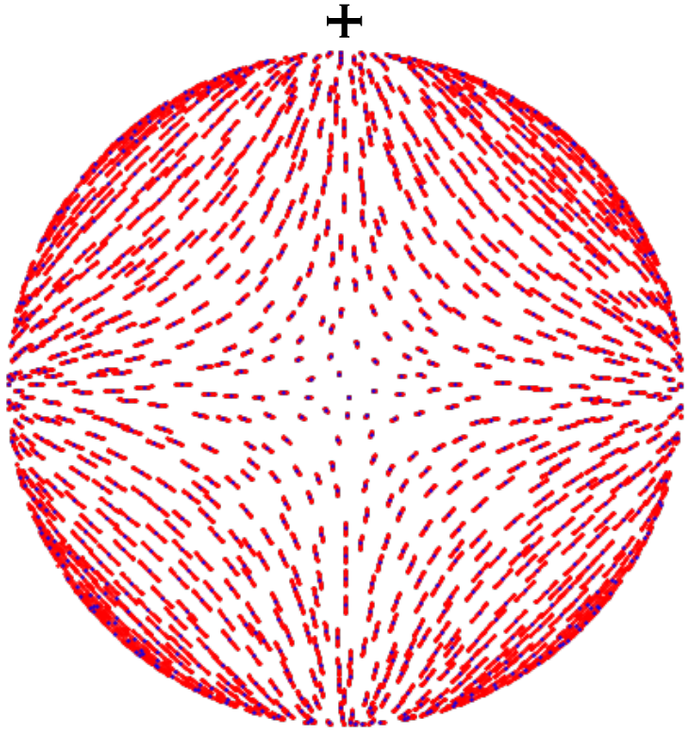


Northern
hemisphere





Southern
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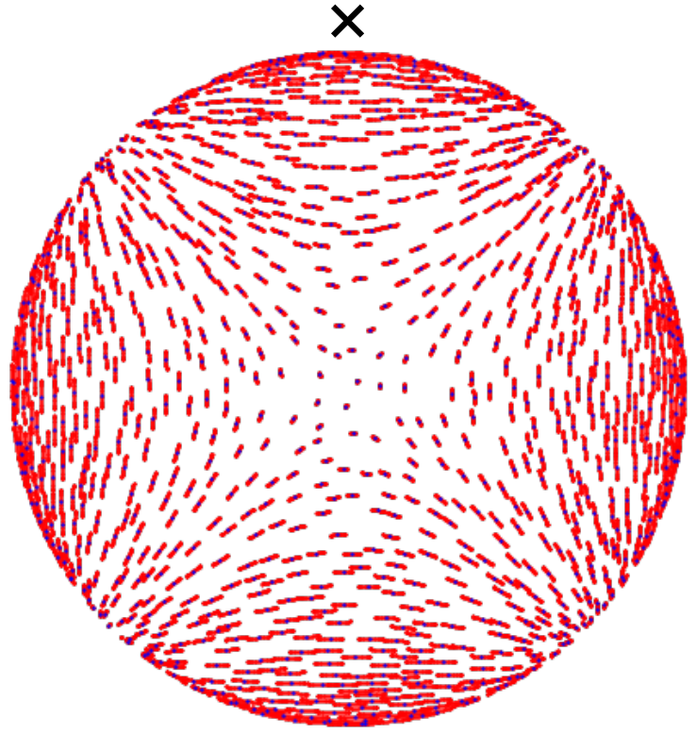
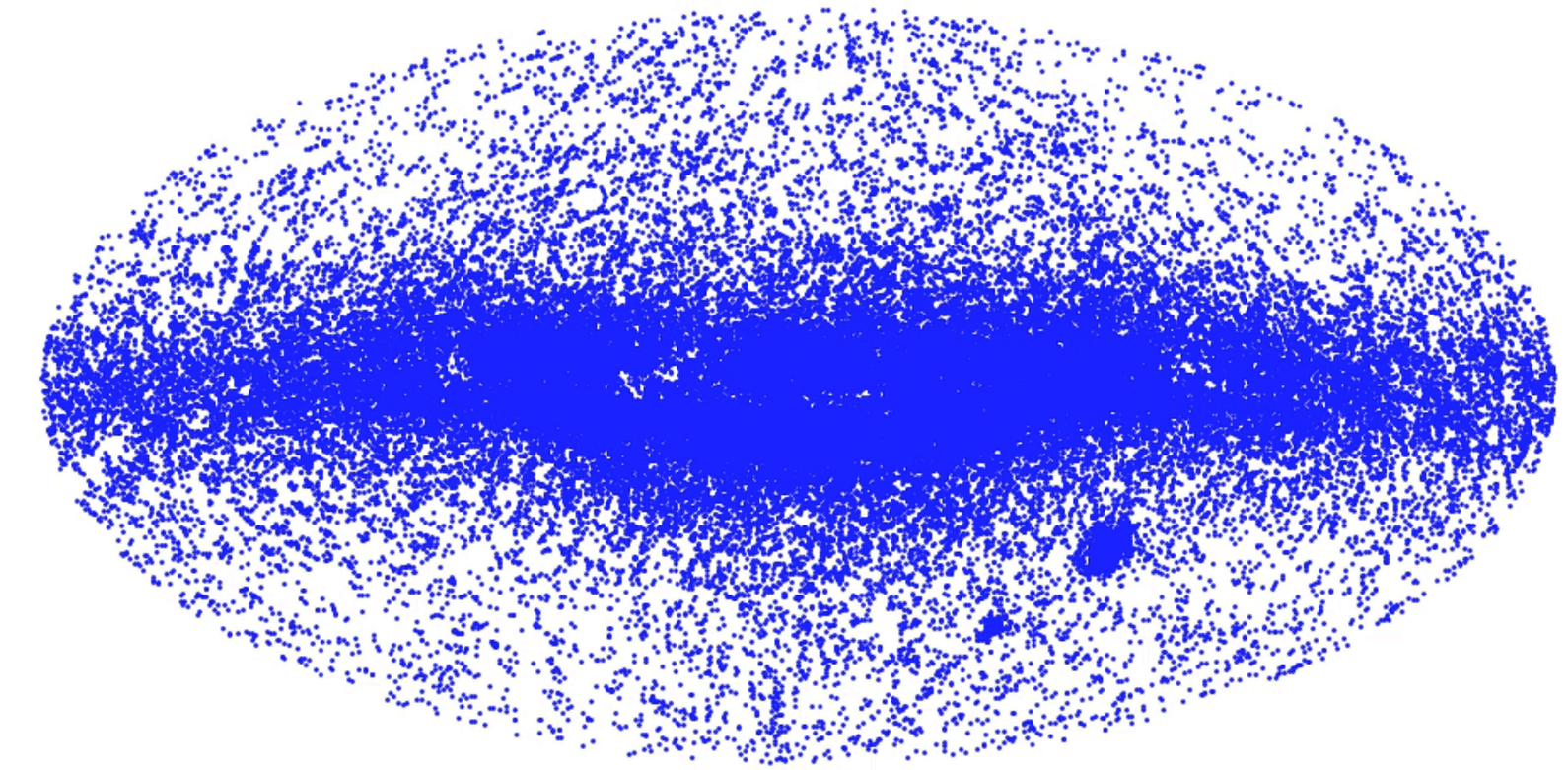
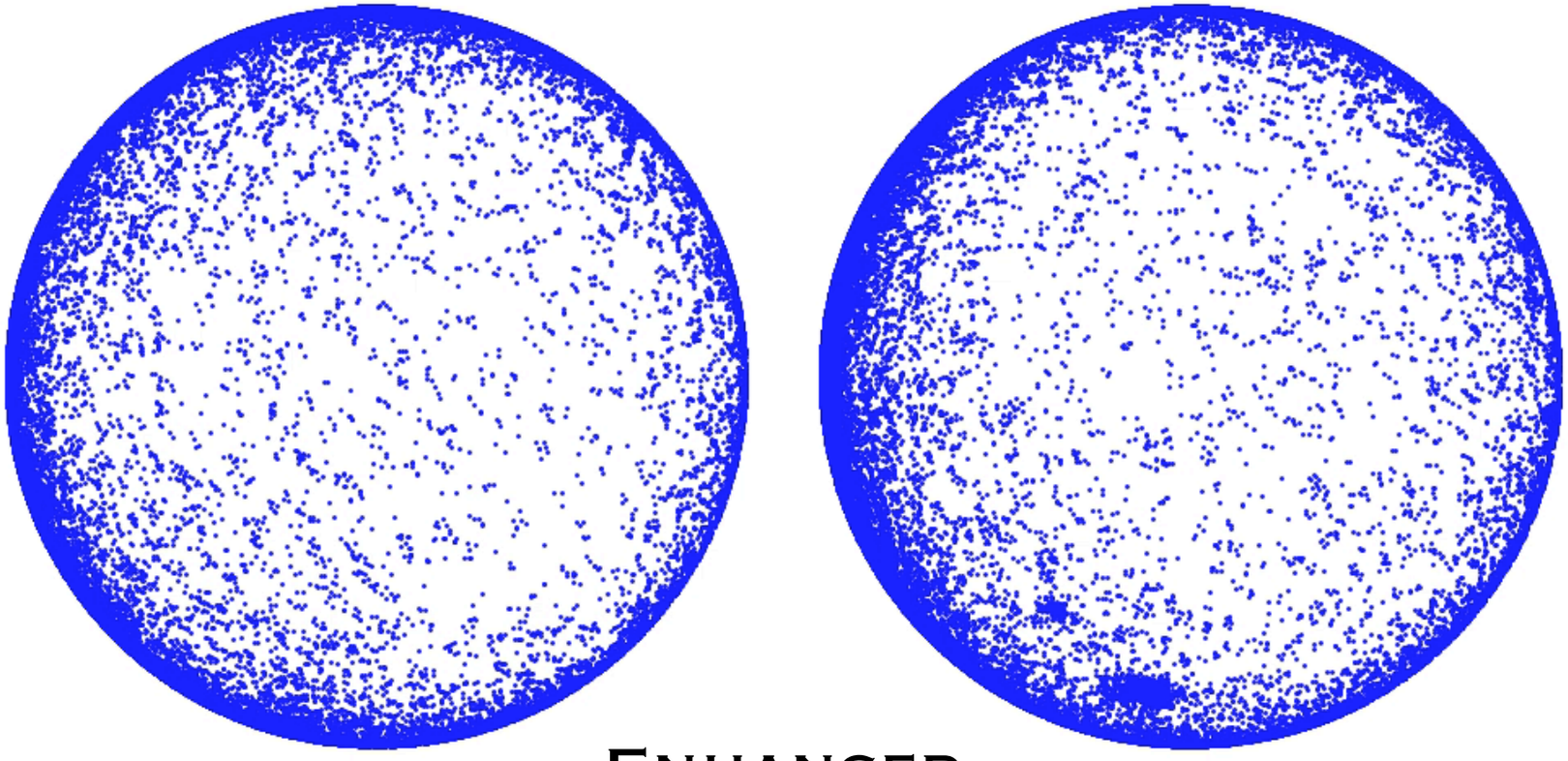


Enhanced
$10^{13}$ times





Enhanced
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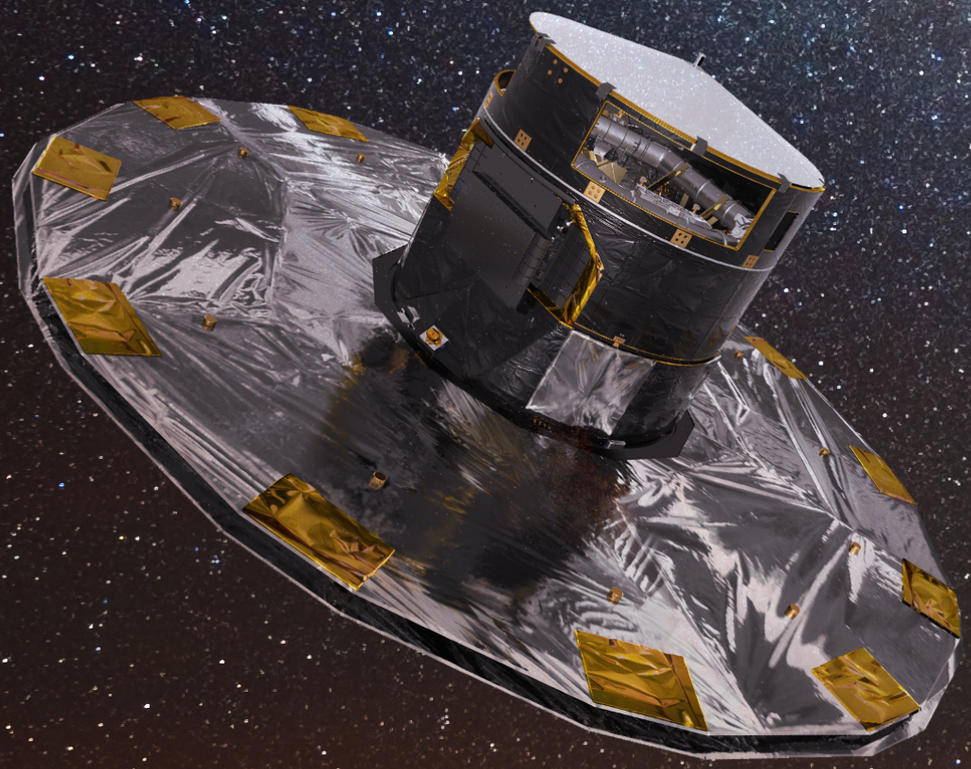
# Is this effect detectable?

# Gaia

## ESA mission for astrometry in the Milky Way

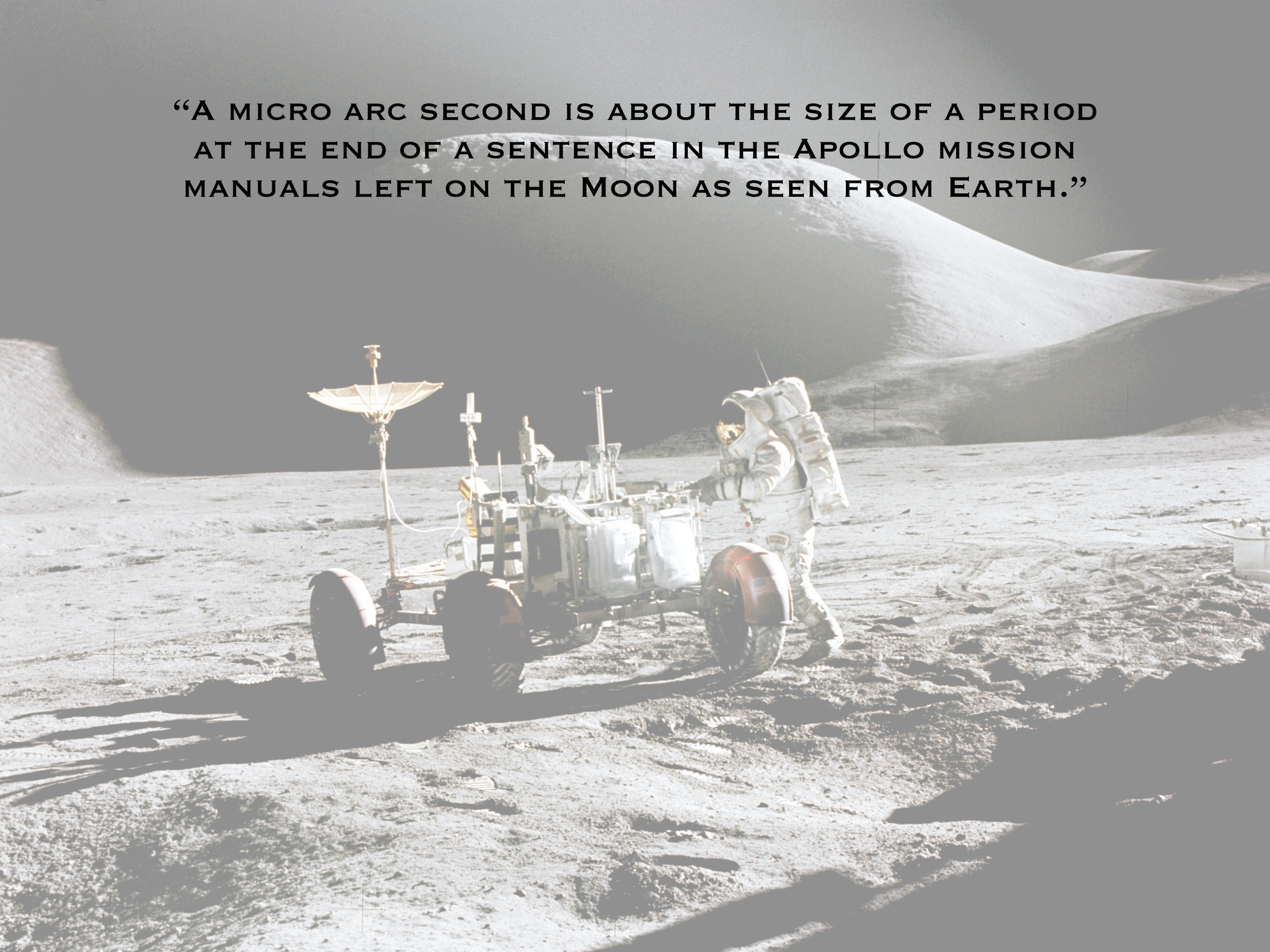


### Objectives:

Map ~$10^9$ objects ~70 times each

Astrometric accuracy
~10 micro arc seconds

"A micro arc second is about the size of a period at the end of a sentence in the Apollo mission manuals left on the Moon as seen from Earth."



# Sensitivity to individual events

~100 measurements of each object

Mission duration 5-10 years

Sensitive in the range $10^{-8}$ - $3\times10^{-7}$ Hz

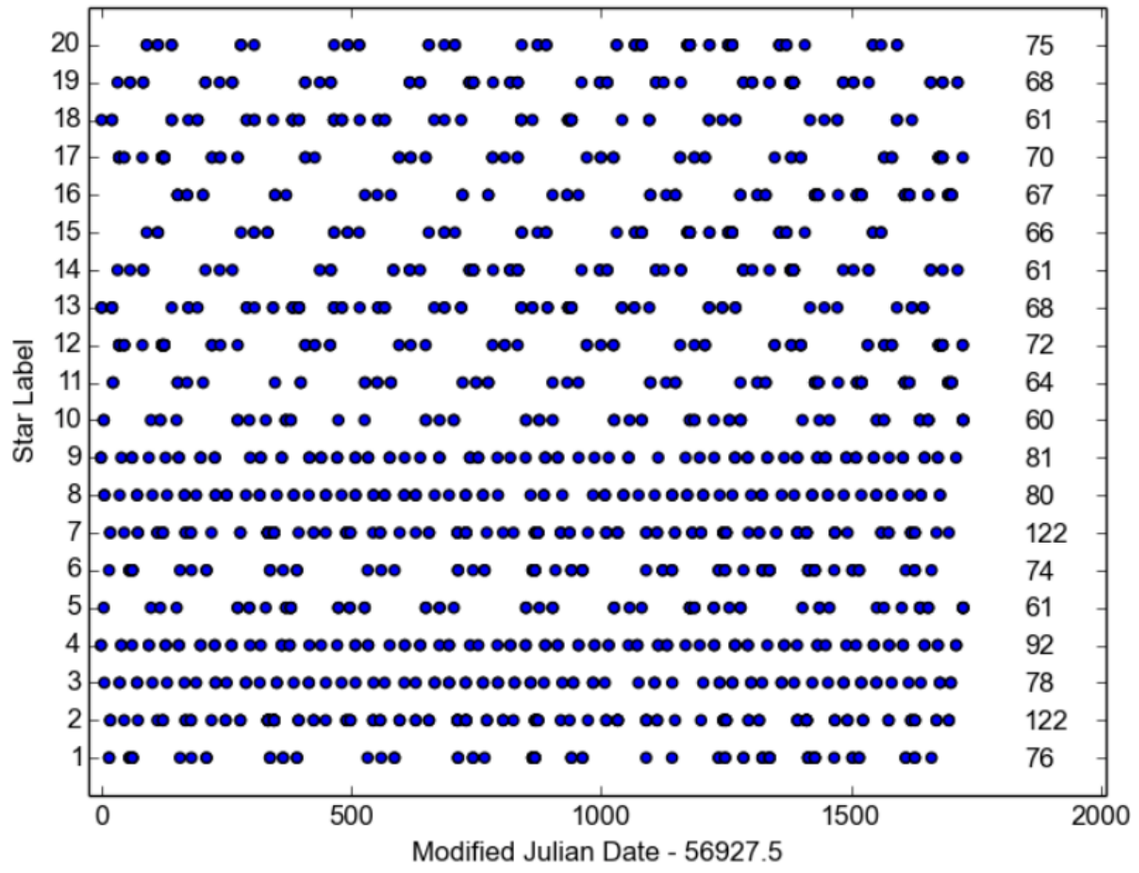
Black hole binaries in the early PN inspiral, $10^{8}$ - $10^{10}$ solar masses

# Prospect for detection

$$\Omega_{\mathrm{GW}} \sim \frac{\sigma^2}{N}$$

# Caveats

## Astrometric measurements are spread over 5 years




## Noise + large scale systematics at this level

## Gaia has a lifetime of 10 years, PTA surveys continually improve

# Detecting GWs with Gaia

It is only being seriously considered now, in the Gaia era

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Biggest challenge is the size of the data set

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Data Releases 1, 2 & 3 do not feature individual astrometric measurements, working with simulated data

# Computational pipeline

## Simulated data

# Computational pipeline

## Simulated data

Proper motion + Noise + GW signal

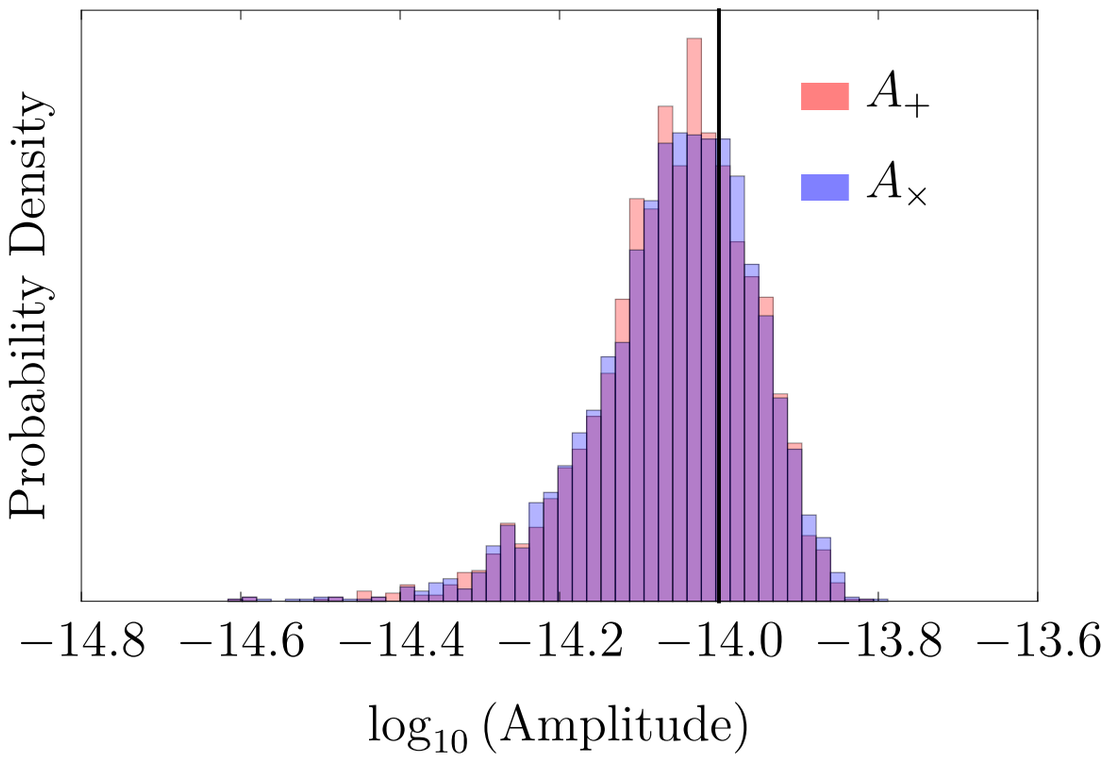
# Computational pipeline

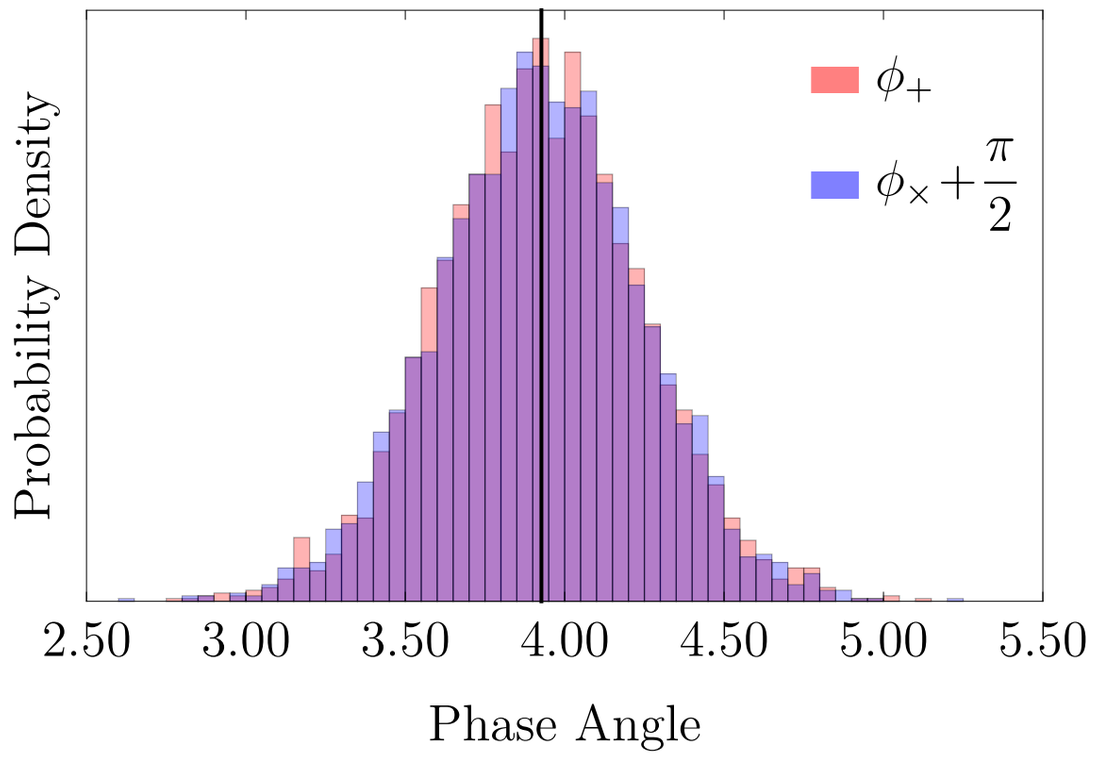
## Simulated data

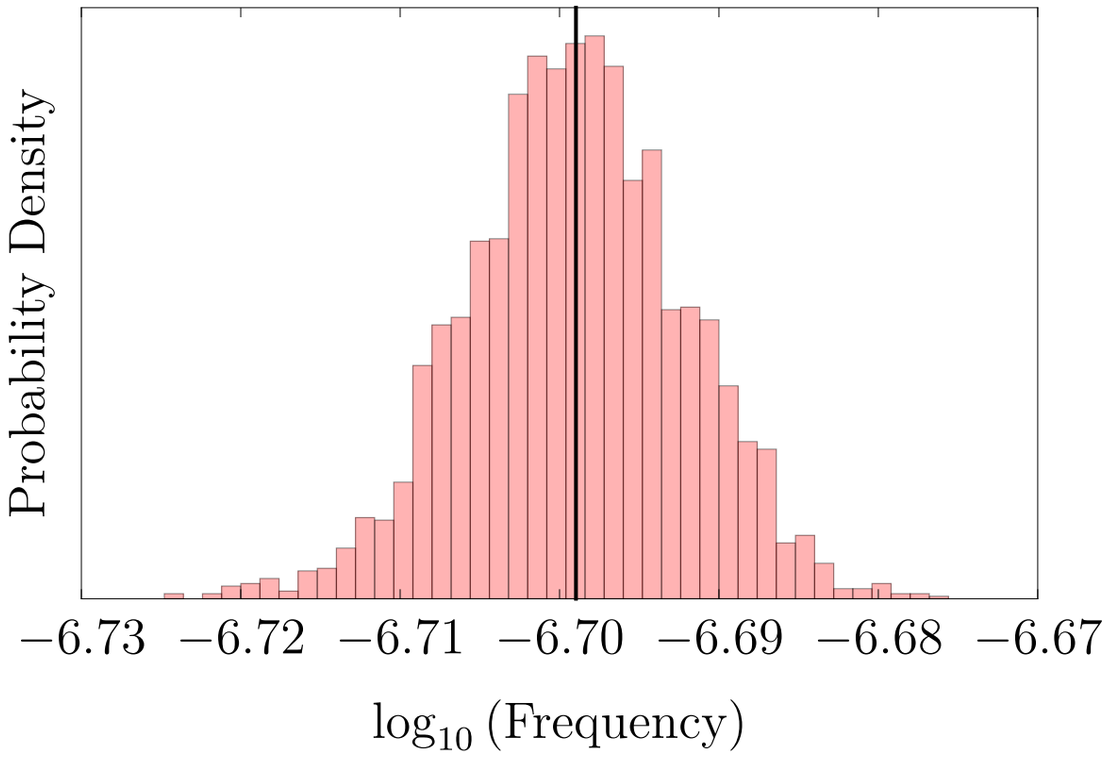
Noise + GW signal

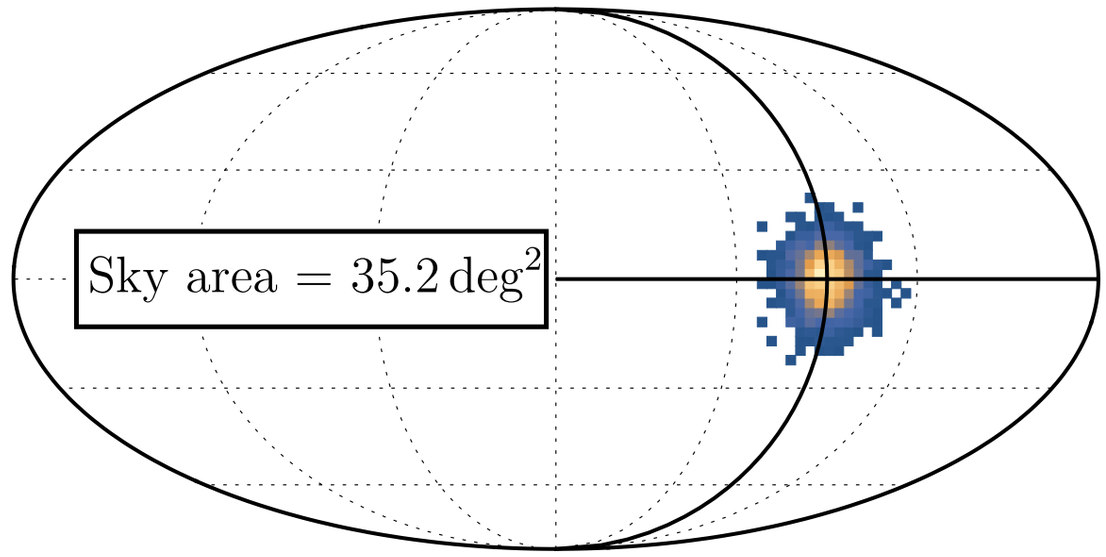
## Pipeline for individual detections:

## Bayesian inference on the parameter grid



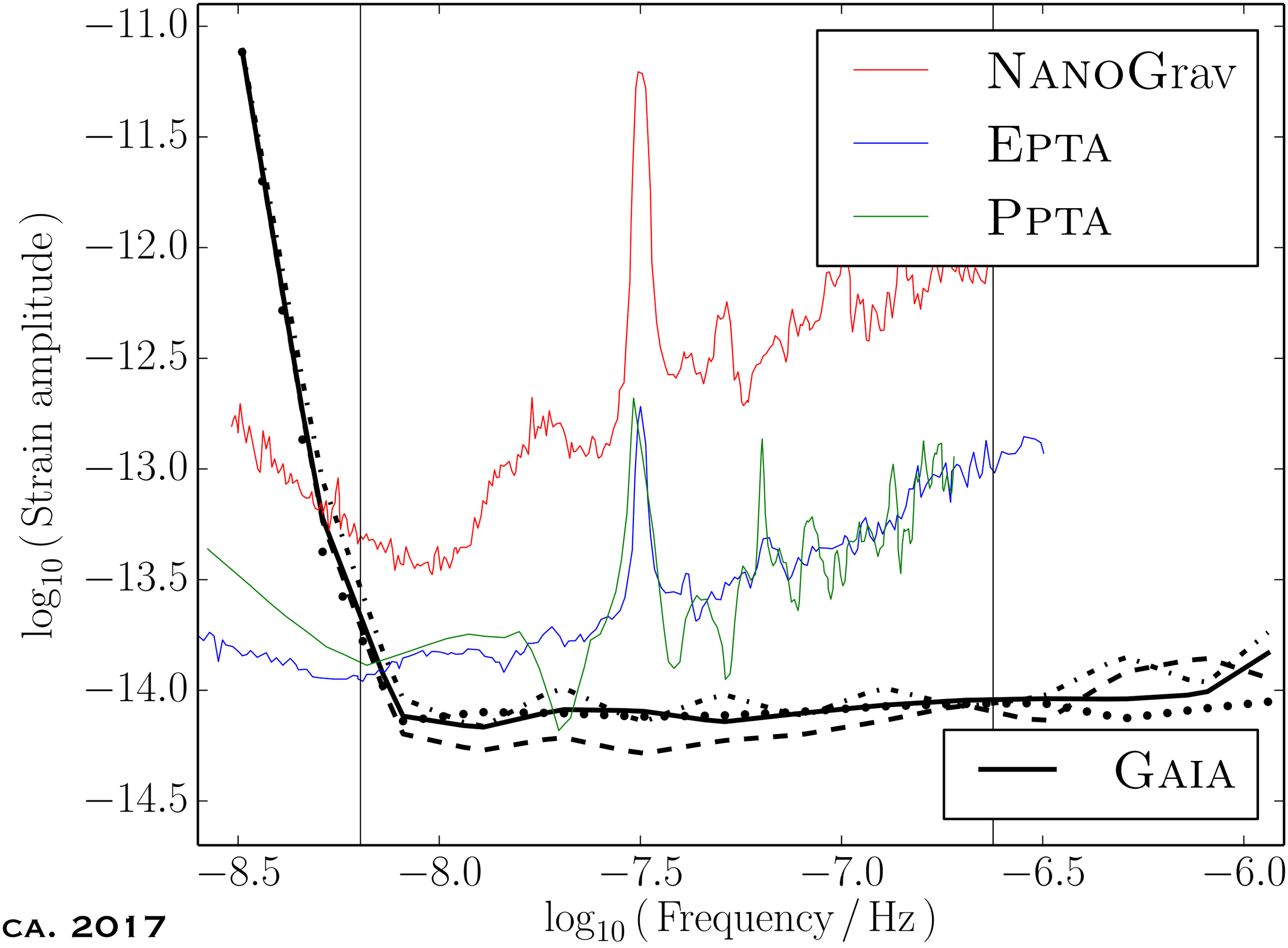



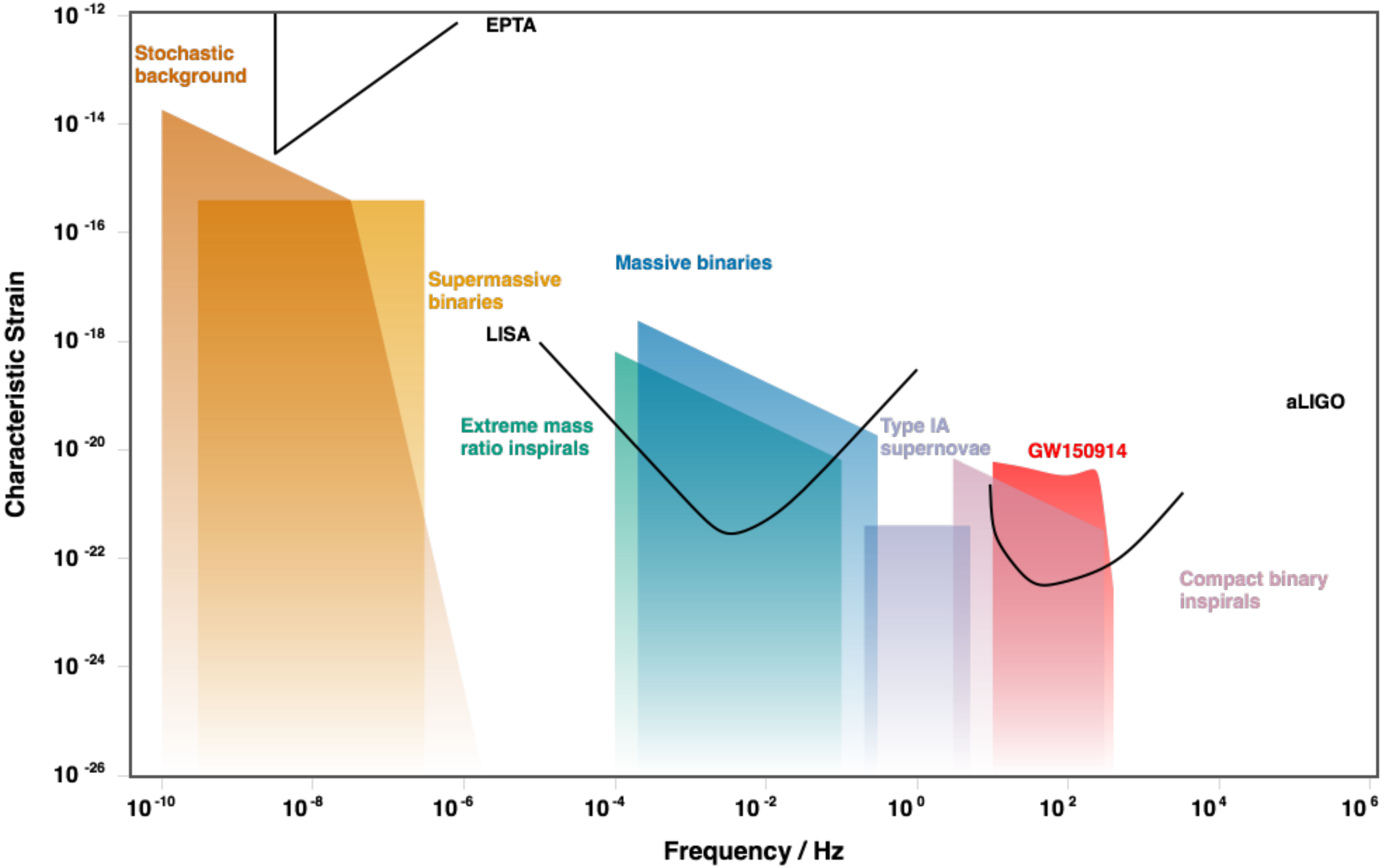




**$10^{10}$ SOLAR MASS BINARY 20 MPC AWAY**

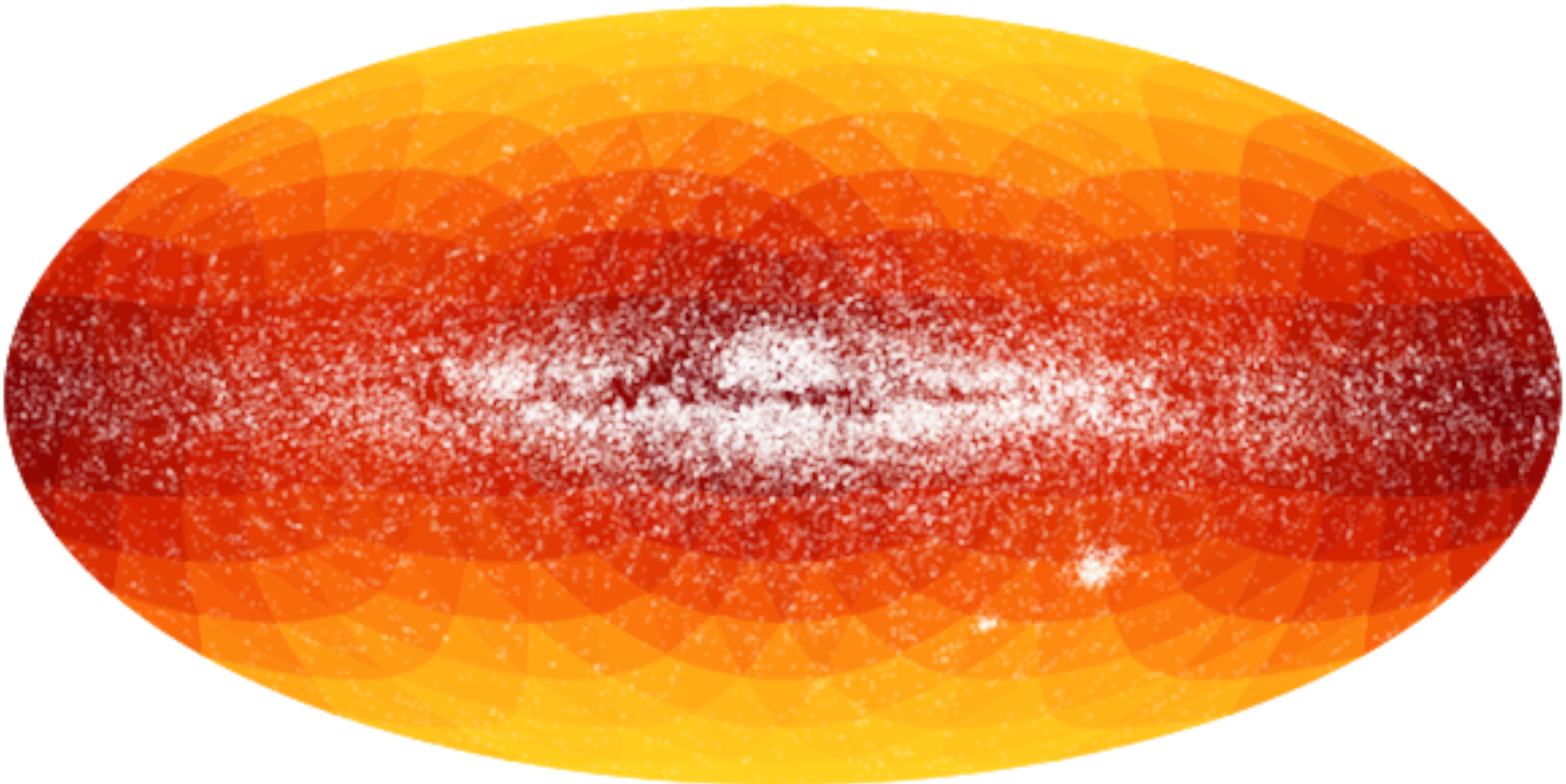
# Frequency Sensitivity of Gaia


NanoGrav
Epta
Ppta
Gaia
−11.0
−11.5
−12.0
−12.5
−13.0
−13.5
−14.0
−14.5
−8.5
−8.0
−7.5
−7.0
−6.5
−6.0
log10 ( Strain amplitude )
log10 ( Frequency / Hz )

ca. 2017


Characteristic Strain
10 -12
10 -14
10 -16
10 -18
10 -20
10 -22
10 -24
10 -26
Stochastic background
EPTA
Supermassive binaries
Massive binaries
LISA
Extreme mass ratio inspirals
Type IA supernovae
aLIGO
GW150914
Compact binary inspirals
10 -10
10 -8
10 -6
10 -4
10 -2
10 0
10 2
10 4
10 6
Frequency / Hz

# Directional sensitivity of Gaia




30% VARIATION ACROSS THE SKY

# Correlations of a Stochastic Background

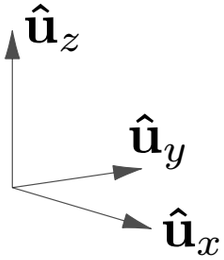
# Stochastic GW background

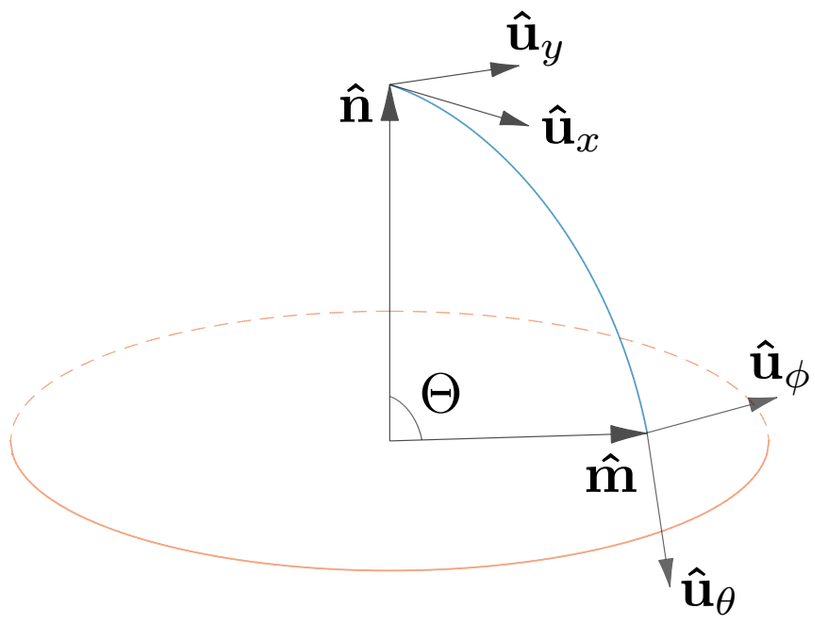
$$\delta n_{\hat{i}} = \frac{n_{\hat{i}} - q_{\hat{i}}}{2(1-\vec{q}\cdot\vec{n})} h_{\hat{j}\hat{k}}(\mathrm{OBS}) n^{\hat{j}} n^{\hat{k}} - \frac{1}{2} h_{\hat{i}\hat{j}}(\mathrm{OBS}) n^{\hat{j}} \,.$$

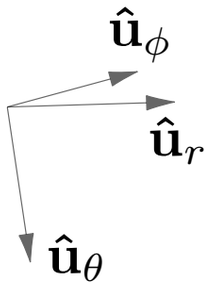
$$h_{ij}(t) = \Re\left\{ \sum_P \int_0^\infty \mathrm{d}f \int_{S^2} \mathrm{d}\Omega_{\mathbf{q}}\, A_P(\mathbf{q}, f)\, e^{-2\pi \mathrm{i} f t} \epsilon_{ij}^P(\mathbf{q}) \right\},$$

# Investigate correlations of stars on the sky

$$\Gamma^{P}_{ij}(\Theta) \propto \int_{S^2} \mathrm{d}\Omega_{\mathbf{q}}\, \delta n_i(n_k, t)\, \delta m_j(m_\ell, t),$$

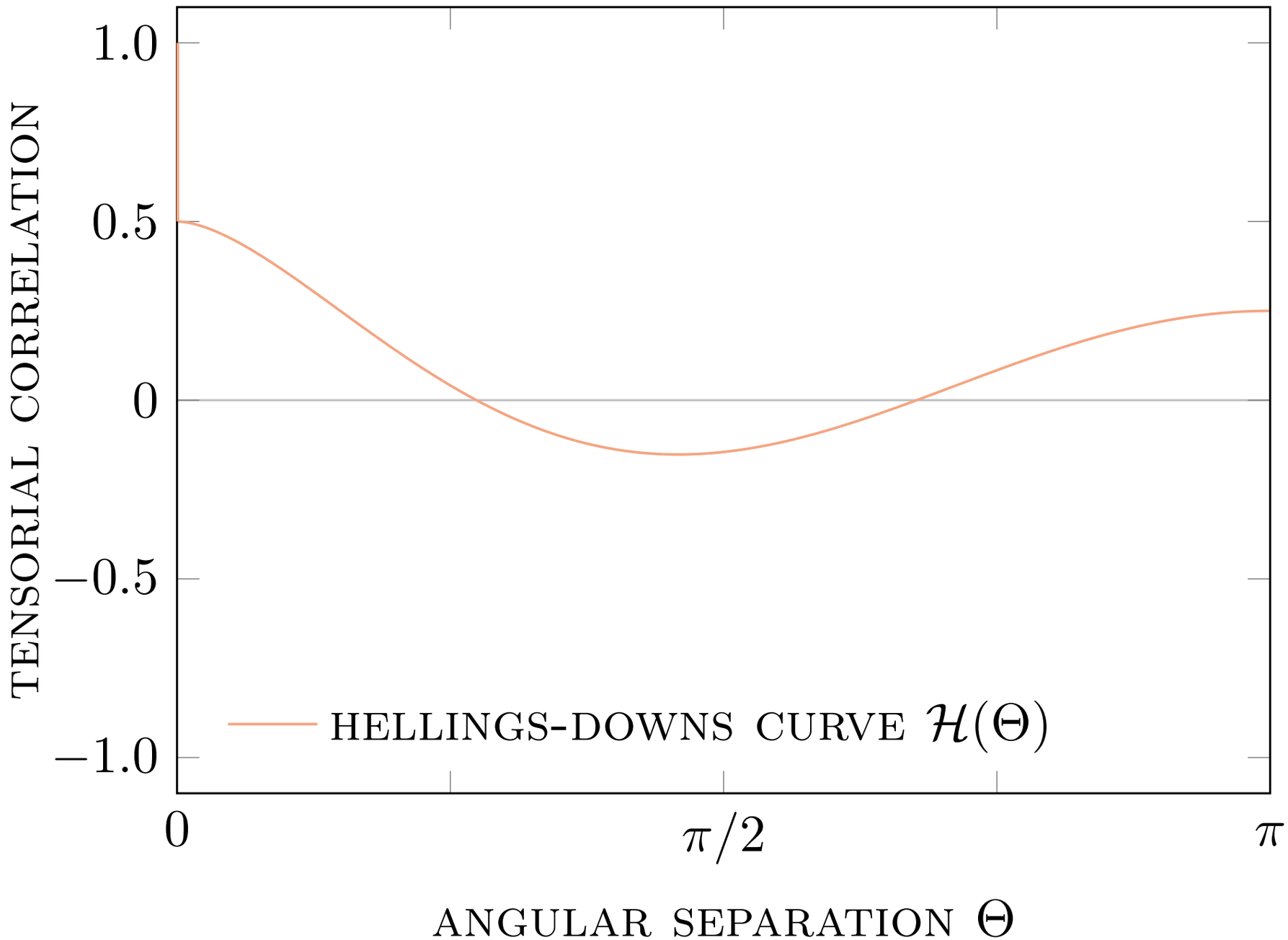


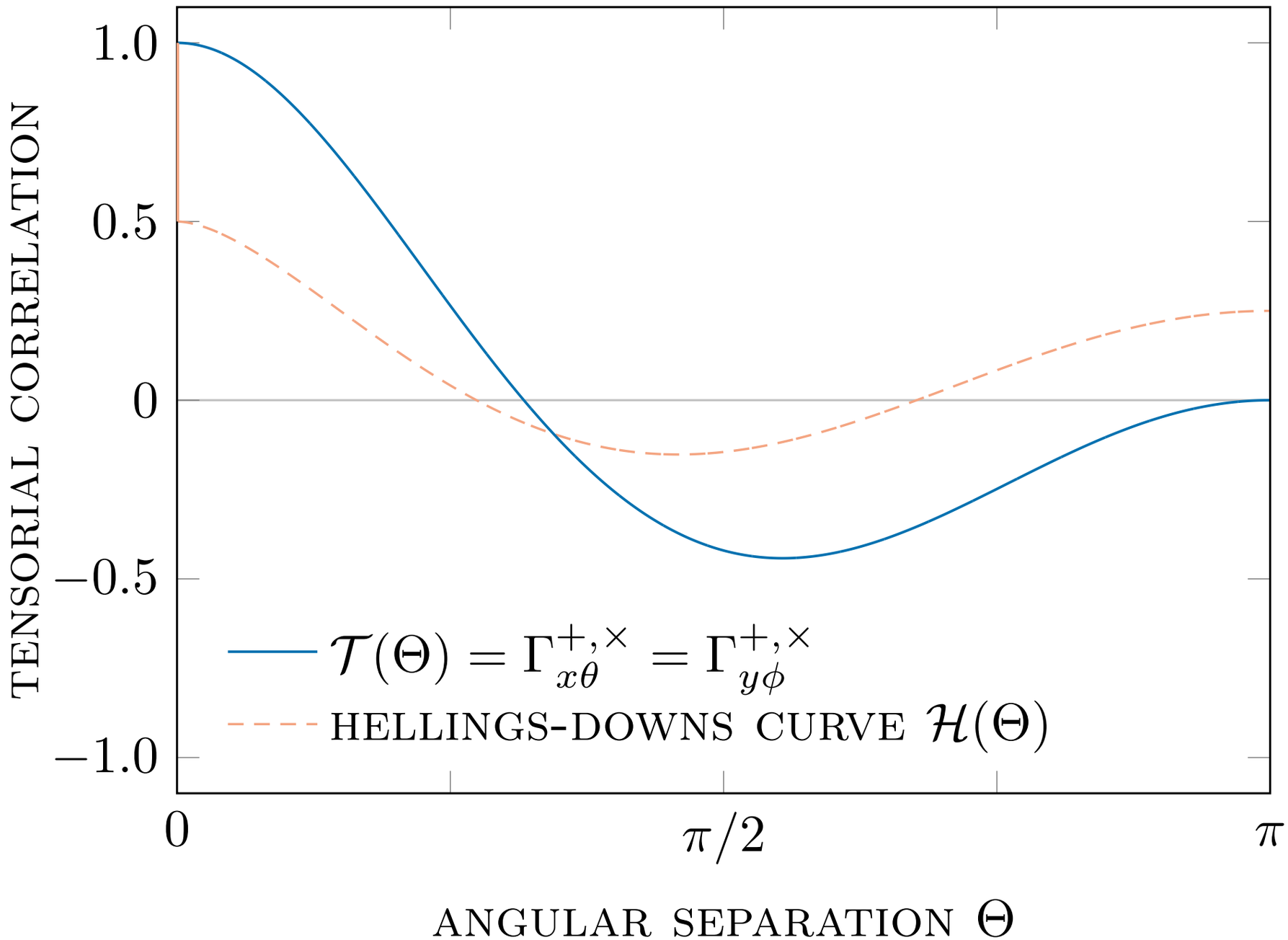

$$C = \left(\begin{array}{c|c} \Gamma_{x\theta} & \Gamma_{x\phi} \\ \hline \Gamma_{y\theta} & \Gamma_{y\phi} \end{array}\right)$$

# GR modes



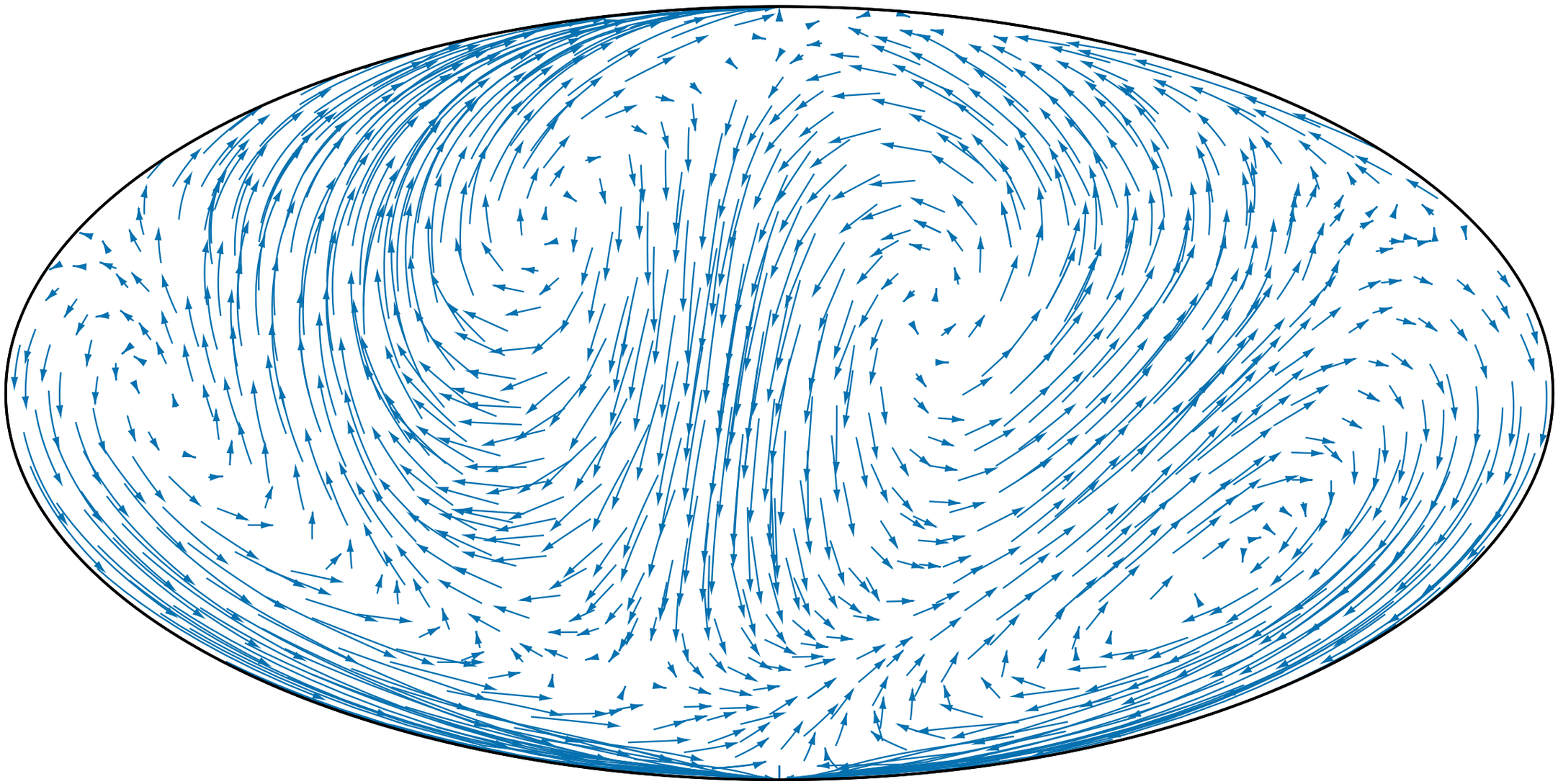

Cf. Book and Flanagan, 2001

# GR modes

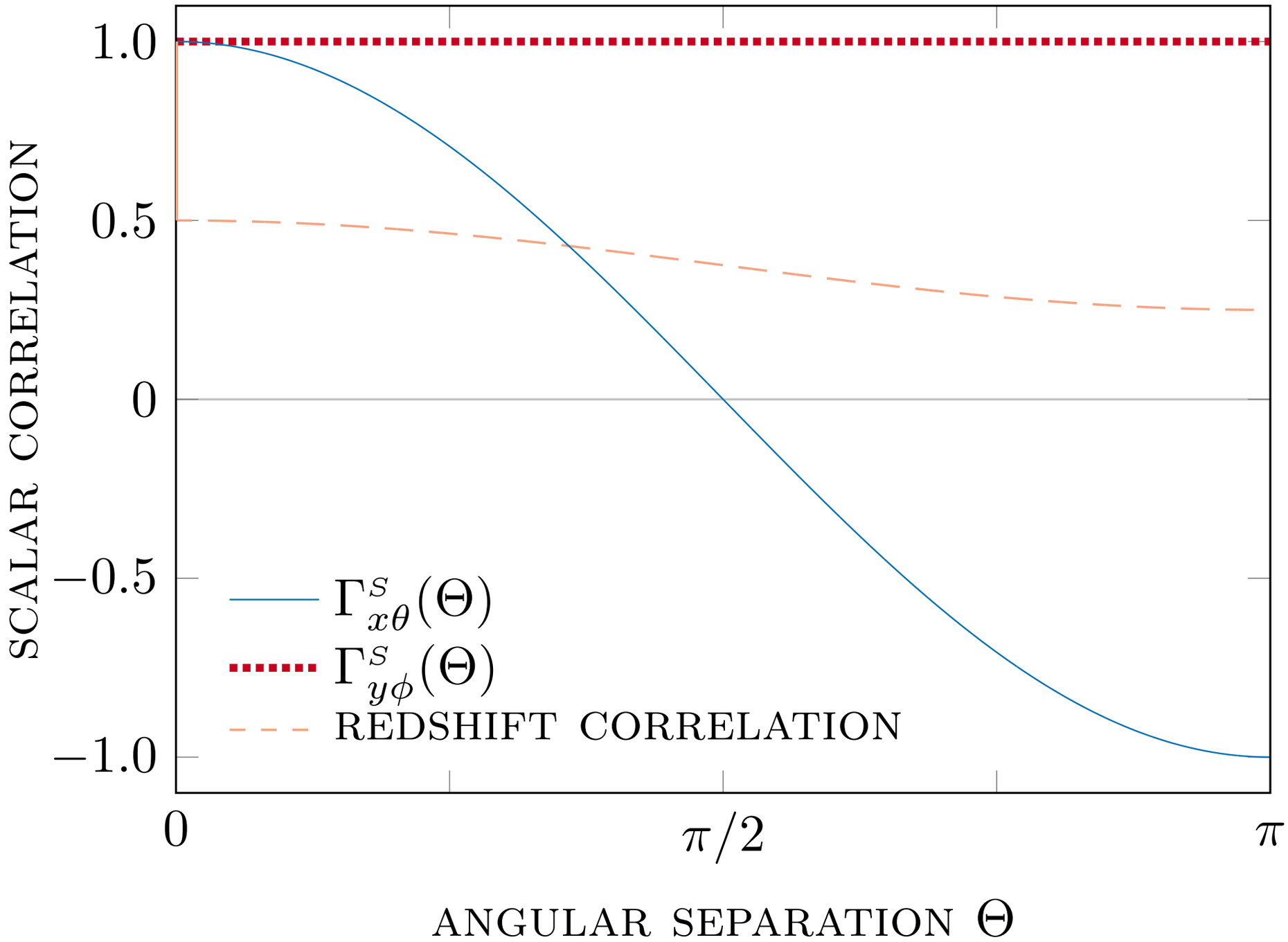



Cf. Book and Flanagan, 2001

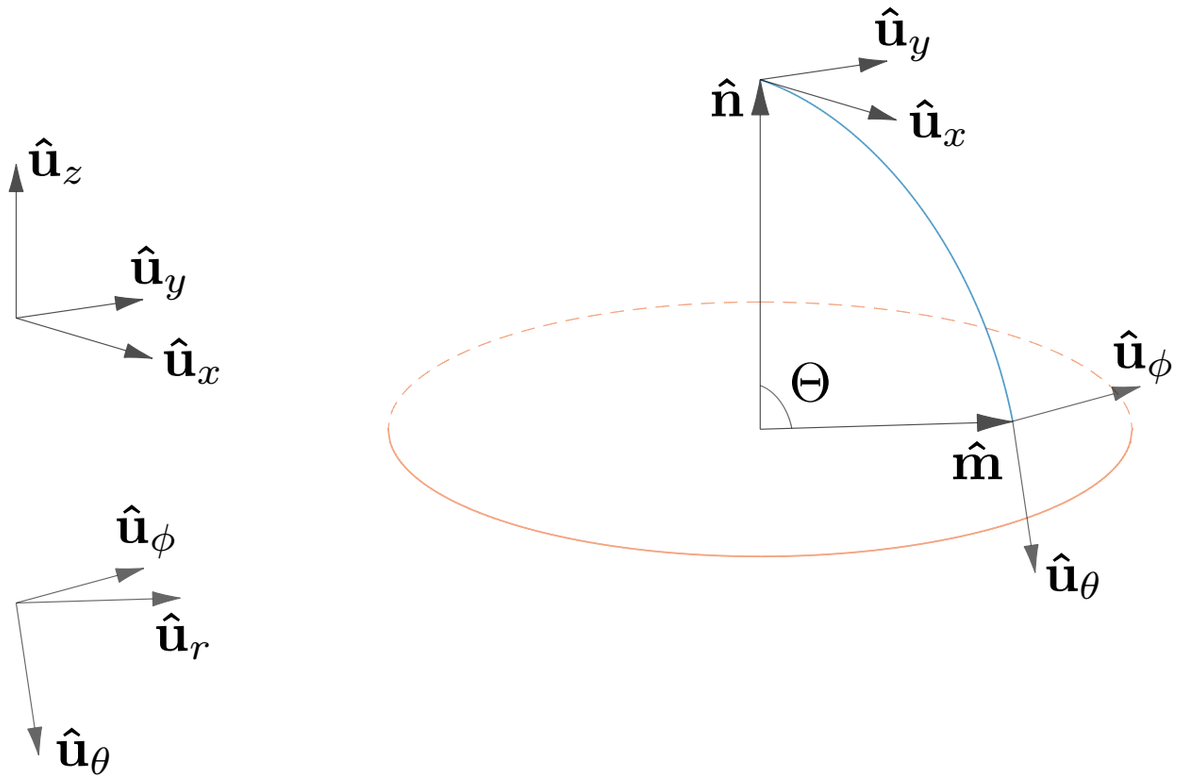
# GR MODES



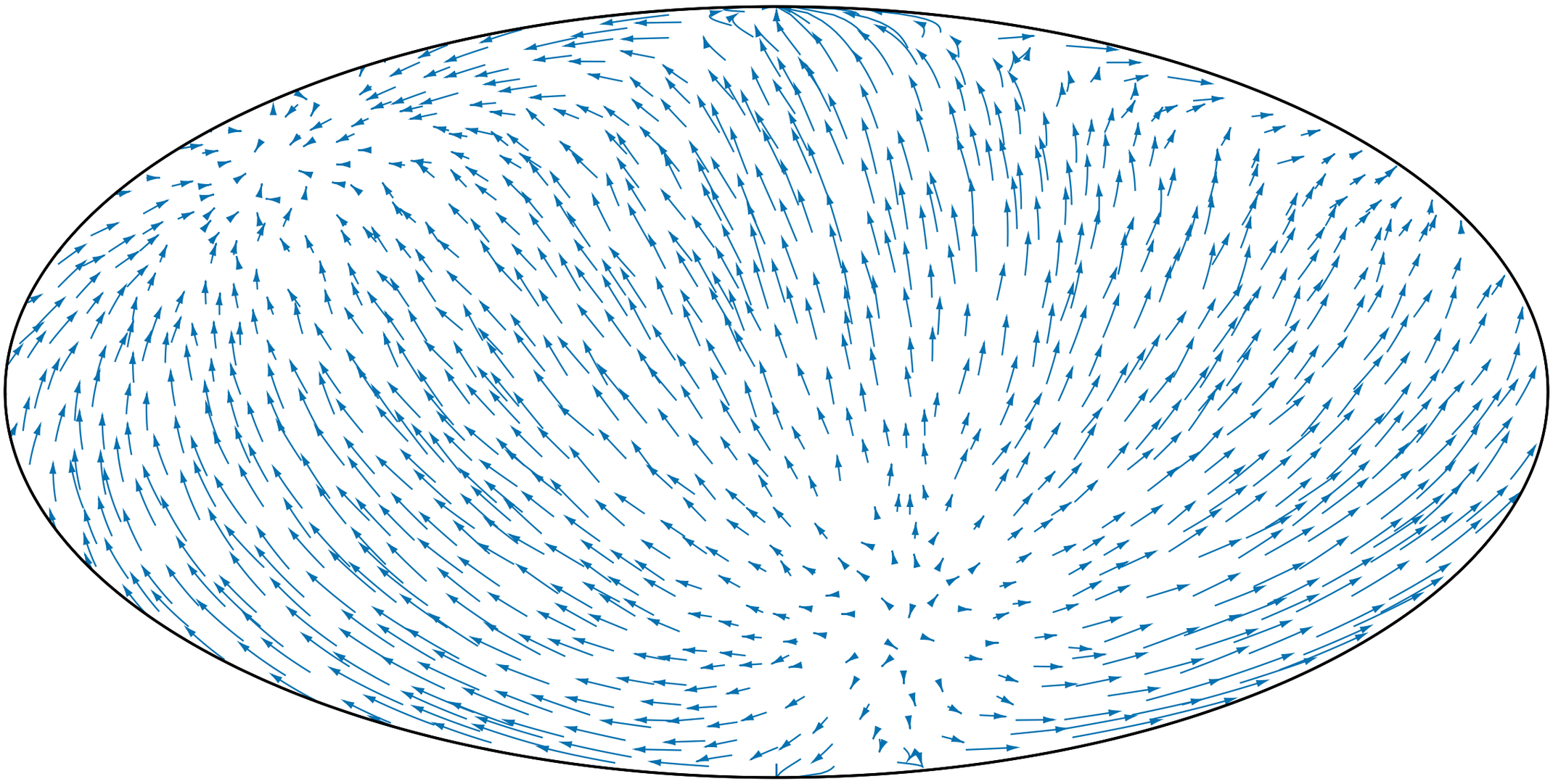
# Breathing mode



# Breathing mode



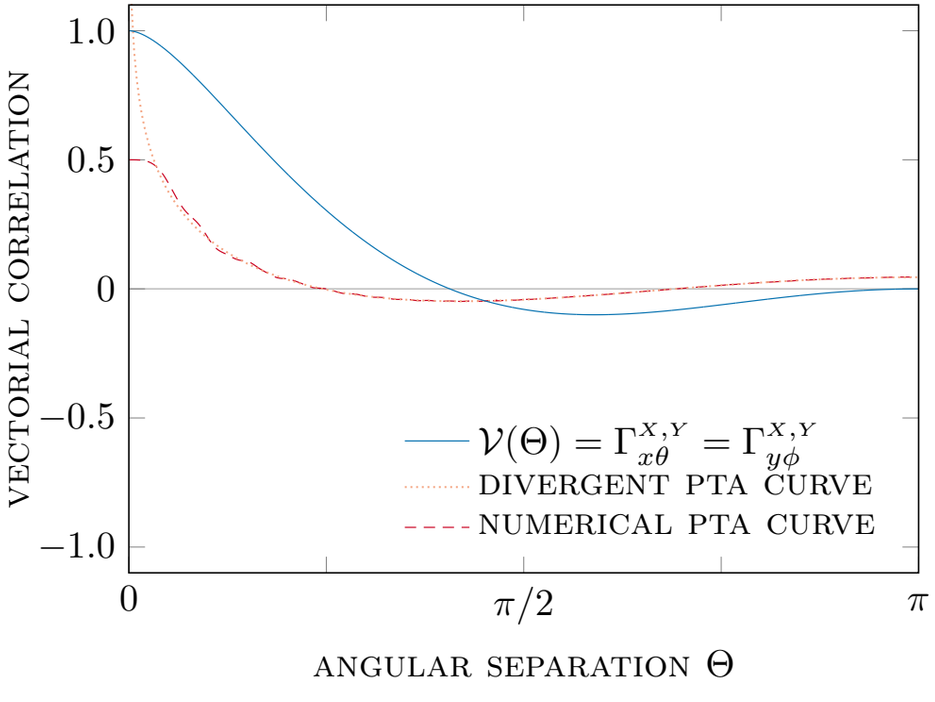
# Breathing mode

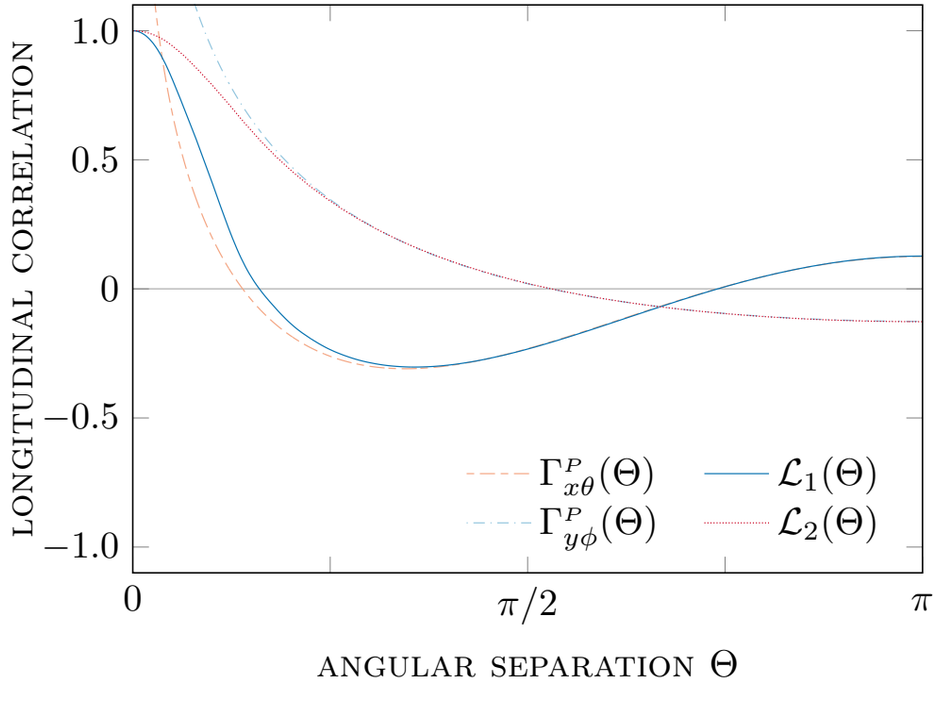


# Vectorial modes

+

# Scalar longitudinal mode

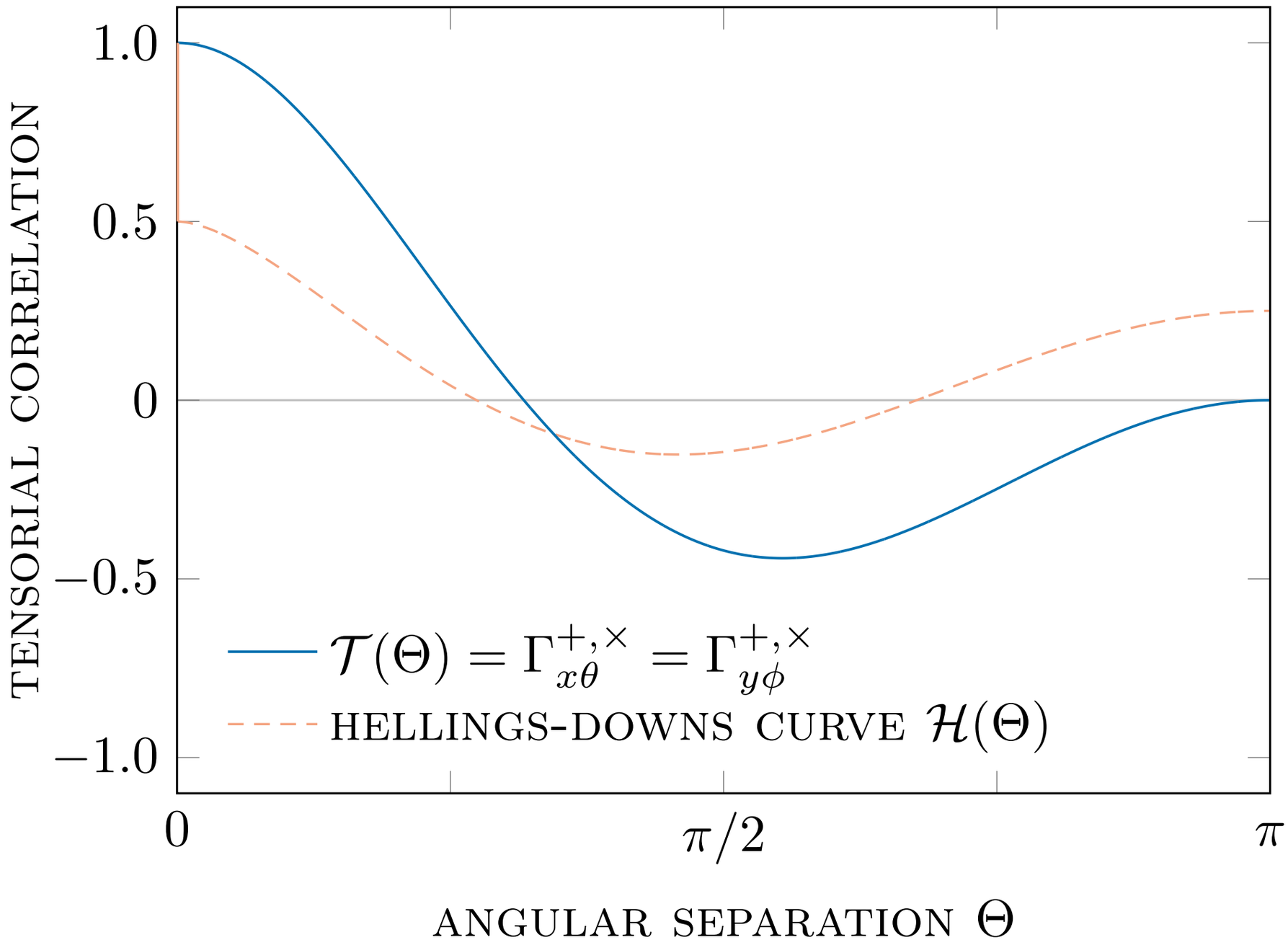



# arXiv:1804.00660

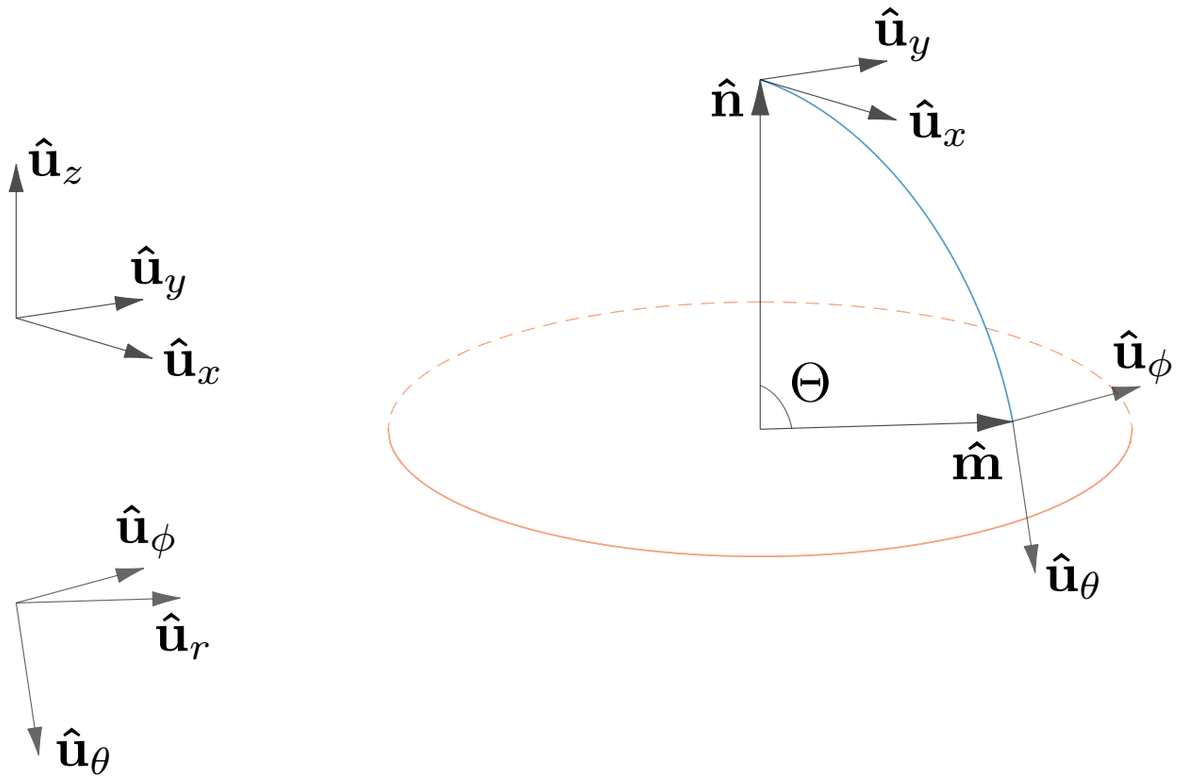
# Redshift-astrometry Correction

# GR modes

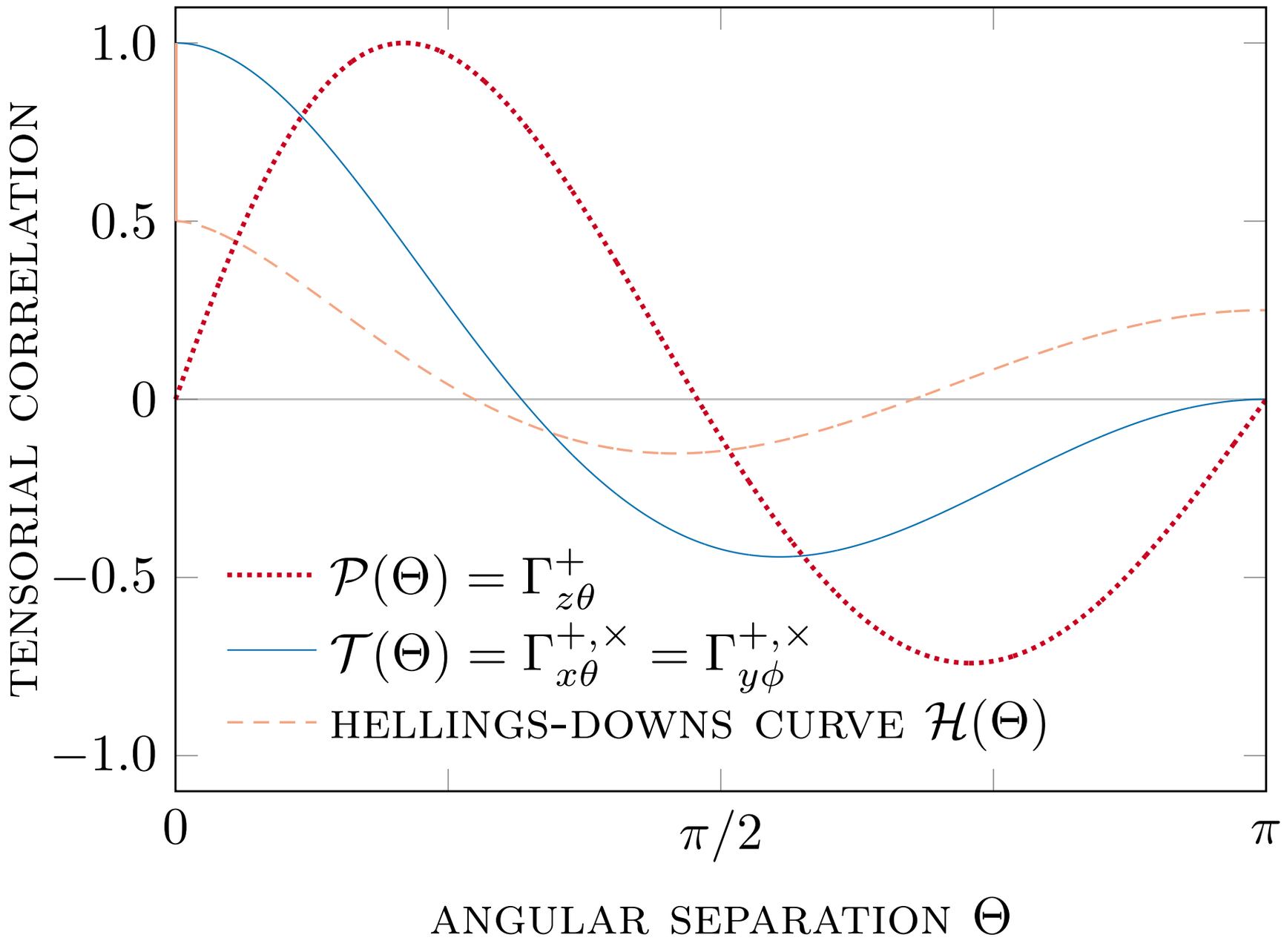



Cf. Book and Flanagan, 2001

# Breathing mode

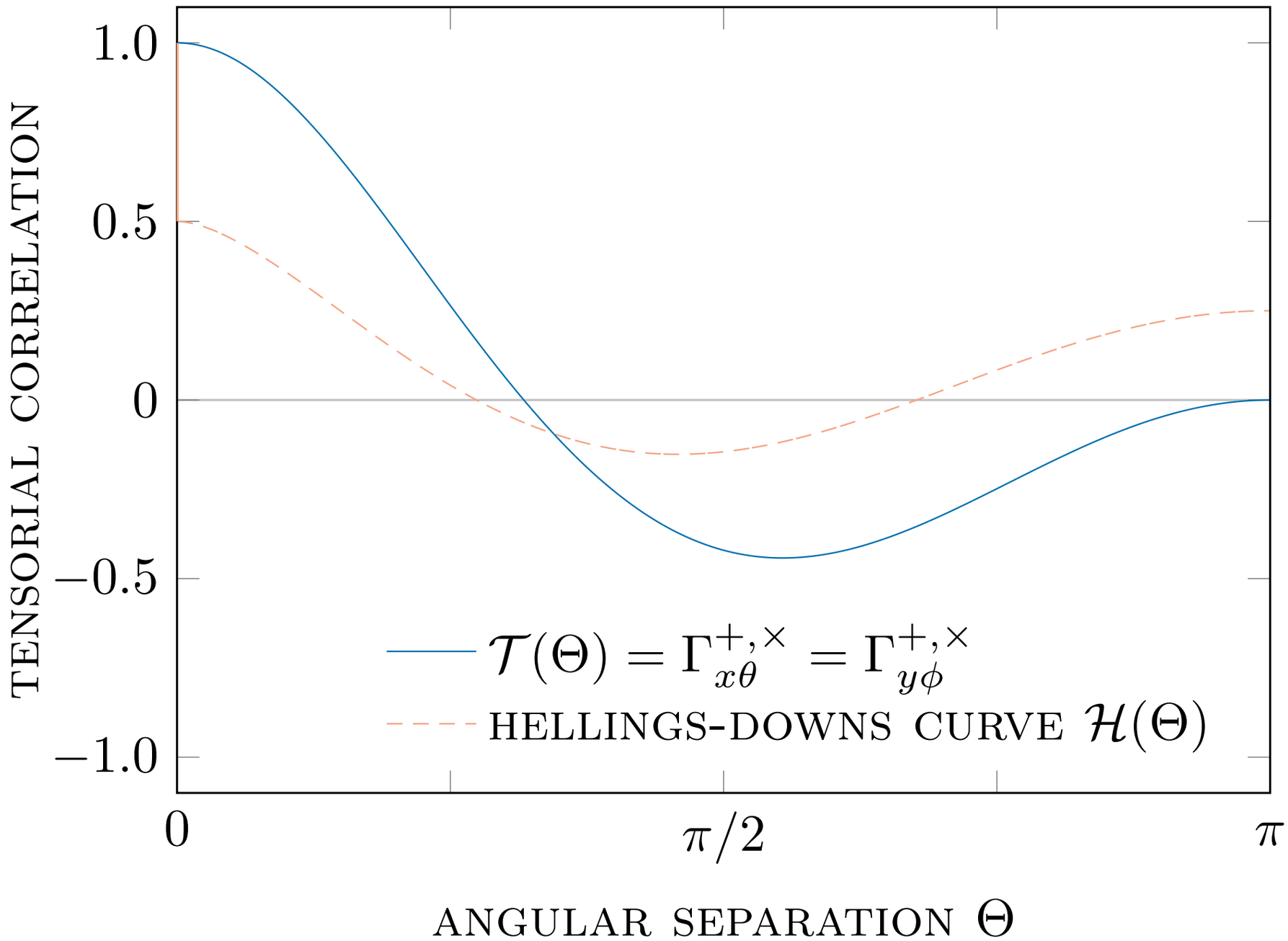


# GR MODES


1.0
0.5
0
−0.5
−1.0
0
π/2
π
TENSORIAL CORRELATION
ANGULAR SEPARATION Θ
$\mathcal{P}(\Theta) = \Gamma^{+}_{z\theta}$
$\mathcal{T}(\Theta) = \Gamma^{+,\times}_{x\theta} = \Gamma^{+,\times}_{y\phi}$
HELLINGS-DOWNS CURVE $\mathcal{H}(\Theta)$

# Massive Graviton Correction

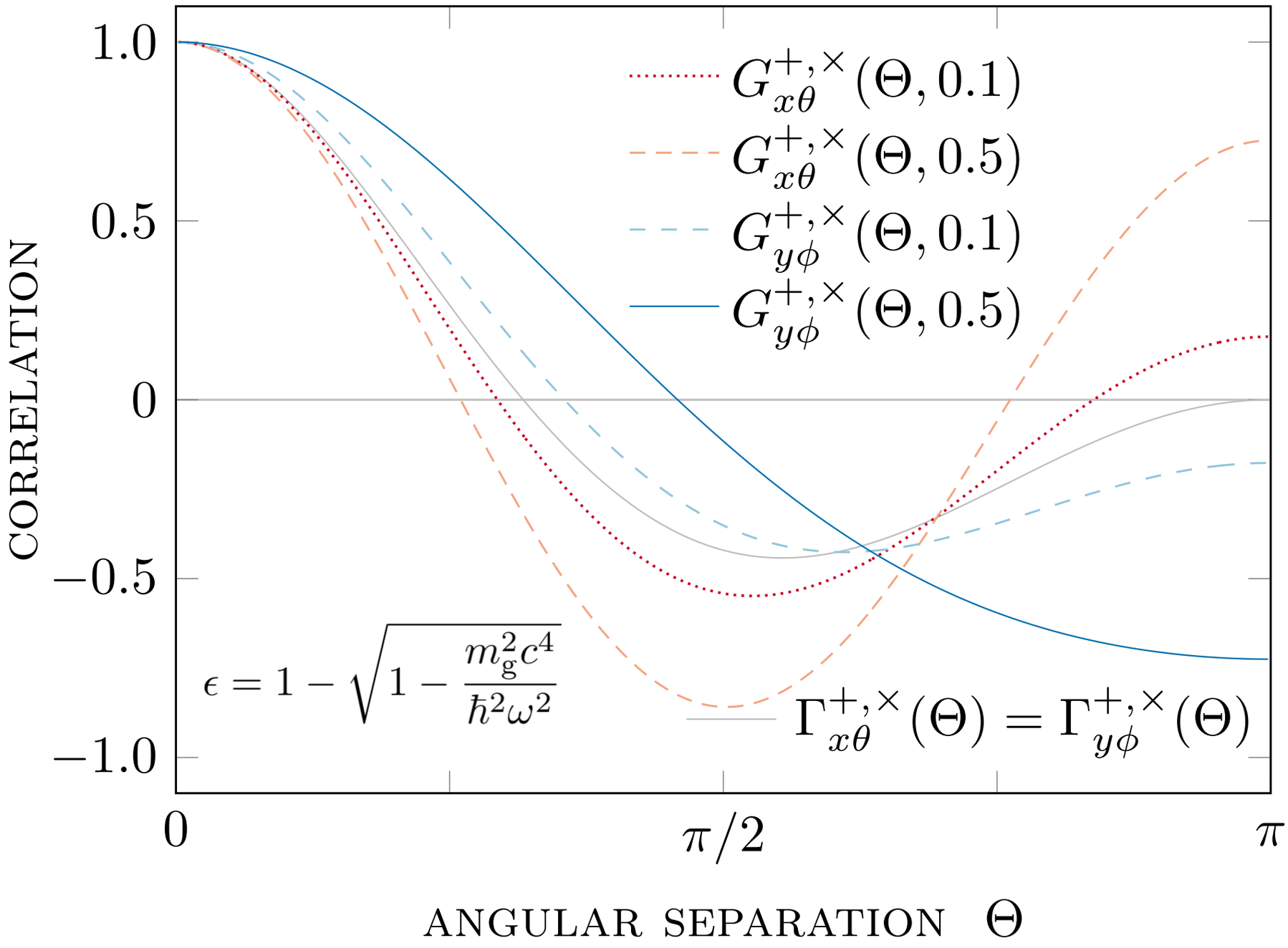
# GR modes




Cf. Book and Flanagan, 2001

# GR MODES WITH MASSIVE GRAVITON CORRECTIONS

$$\Gamma^{\mathrm{P}}_{ij}(\Theta, \epsilon)$$



# Conclusions

1. GWs induce periodic perturbations in the astrometric measurements of stars

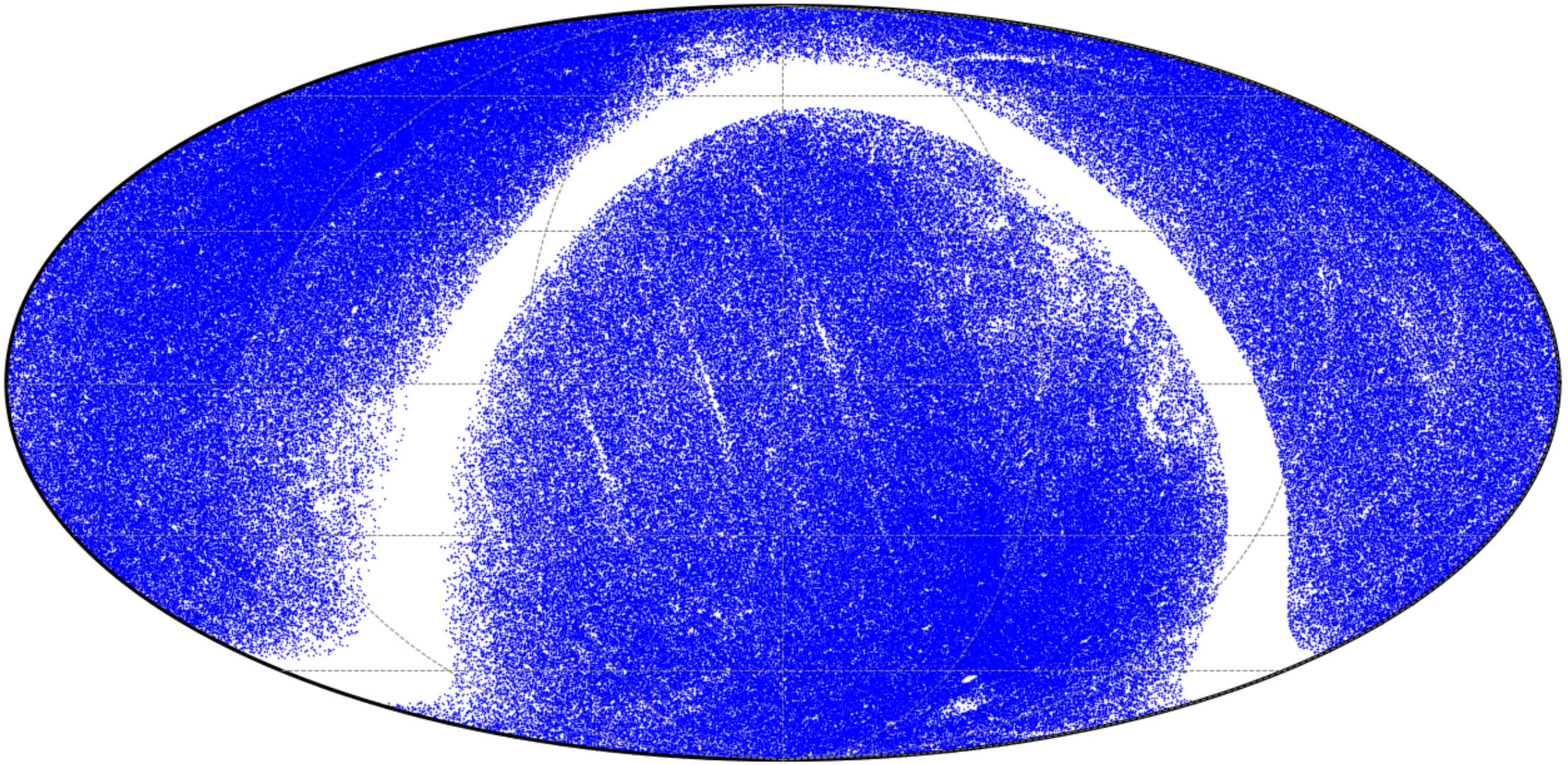
2. Gaia is the ideal tool to study this effect

3. We have developed a new data analysis pipeline

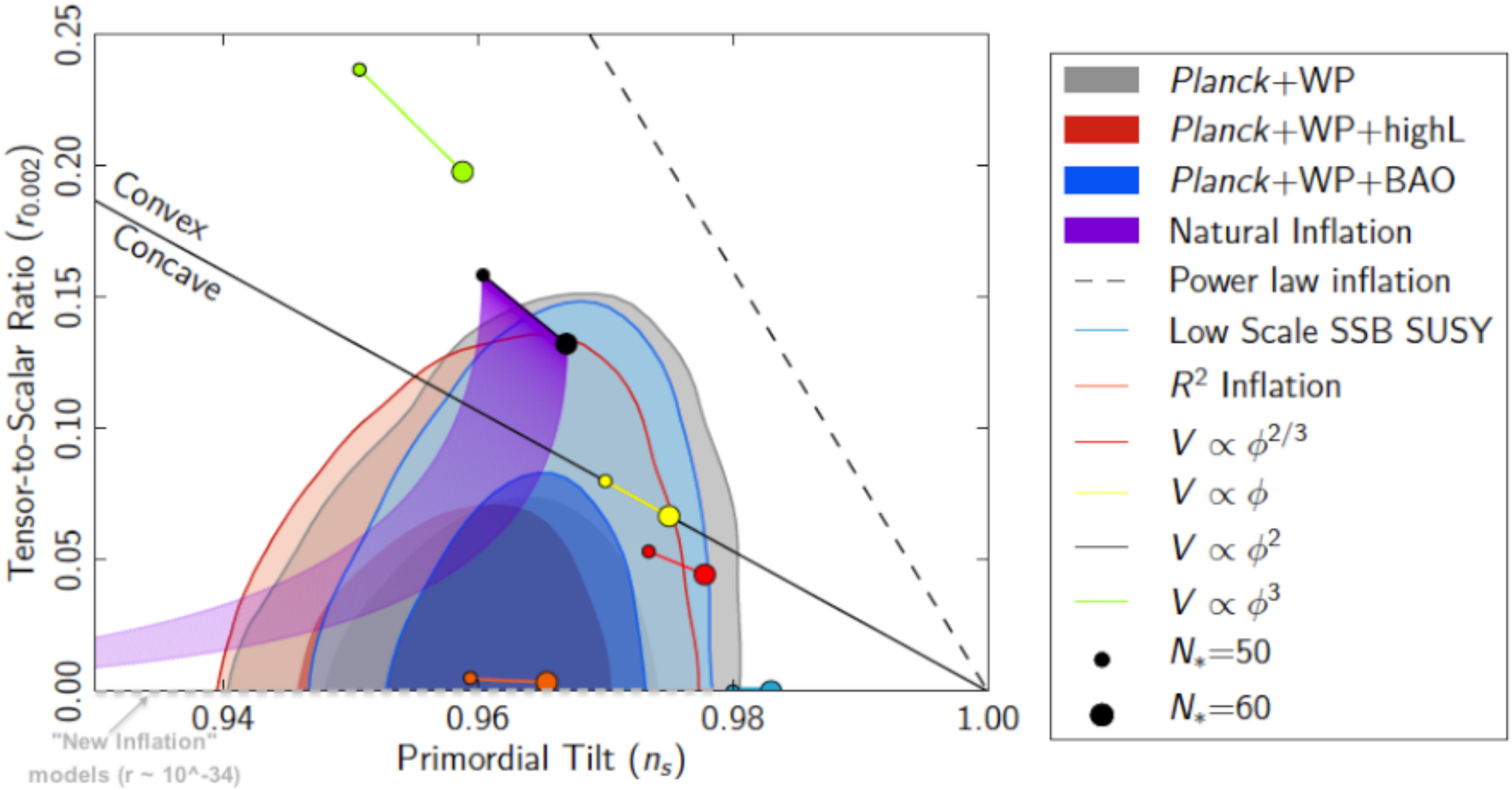
5. Data can be compresses with little loss

6. The final data release will allow GW searches.

# Further work



# Further work



# Further work

1. Systematics in Gaia DR4?
2. Cross-correlation with PTAs
3. Anisotropy tests
4. Your ideas?
5. Awarded grant for studying this!

# BULGARIA

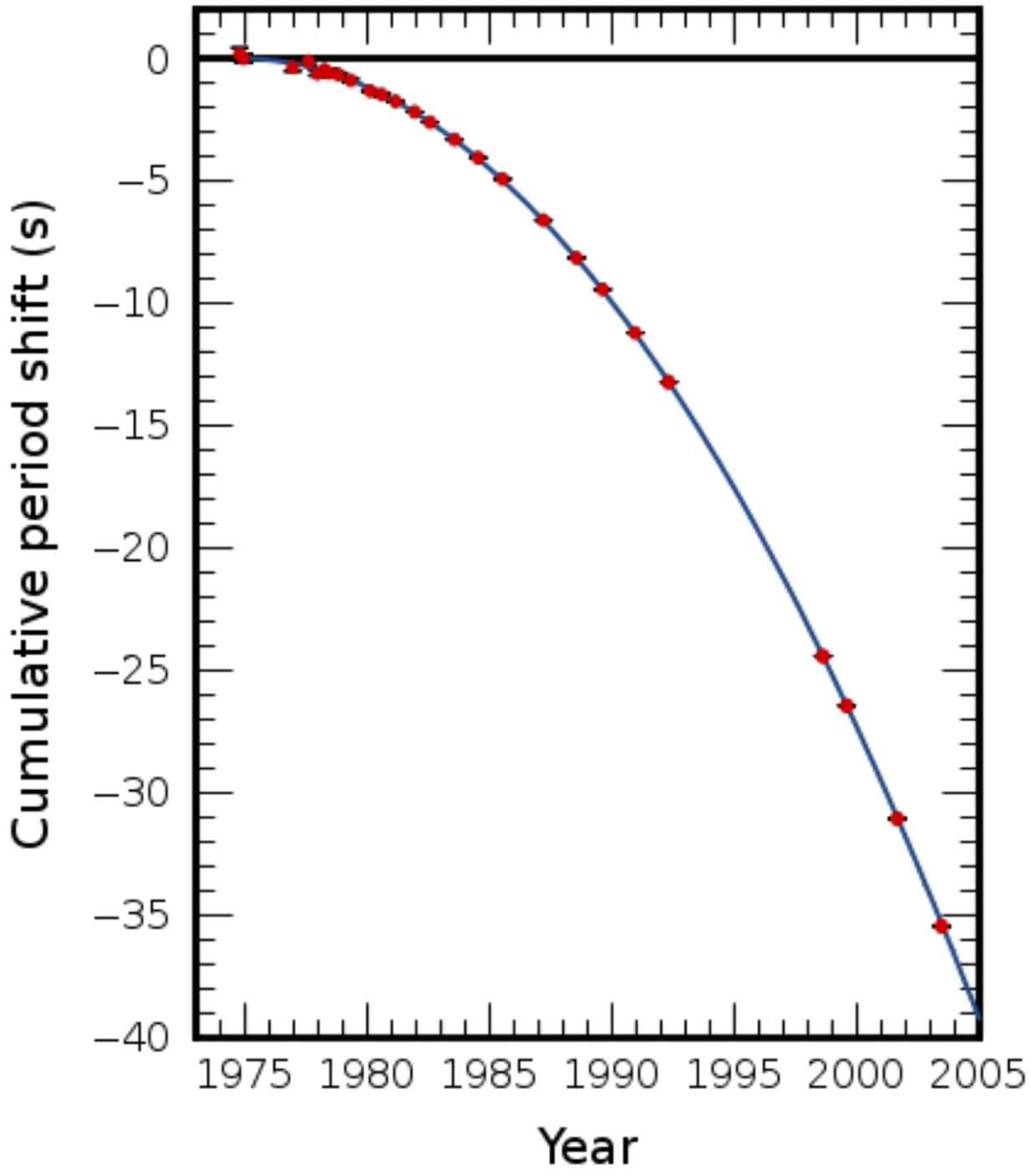
| Score | Rank | Score change |
| --- | --- | --- |
| **43**/100 | **76**/180 | **-2** since 2023 |

CASE WESTERN RESERVE UNIVERSITY

Max-Planck-Institut
für Gravitationsphysik
Albert-Einstein-Institut

gaia

# Hulse-Taylor Binary Pulsar

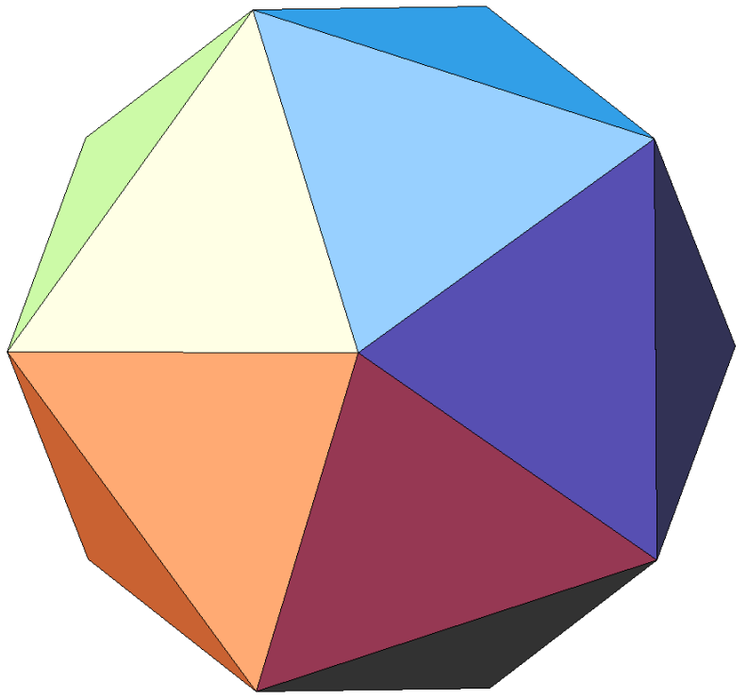


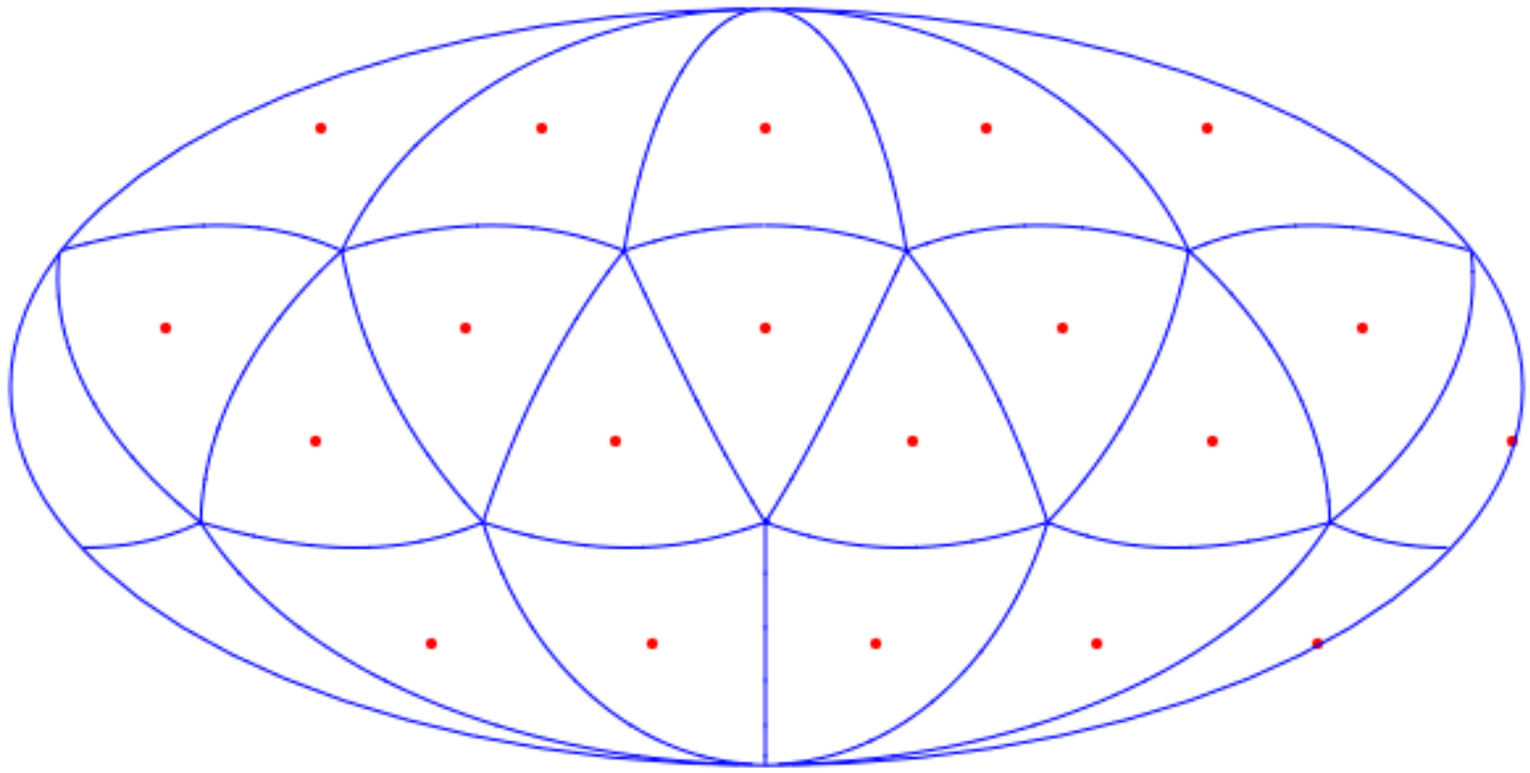

- First binary pulsar discovered in 1974
- Period decay consistent with GR
- Nobel prize in 1994

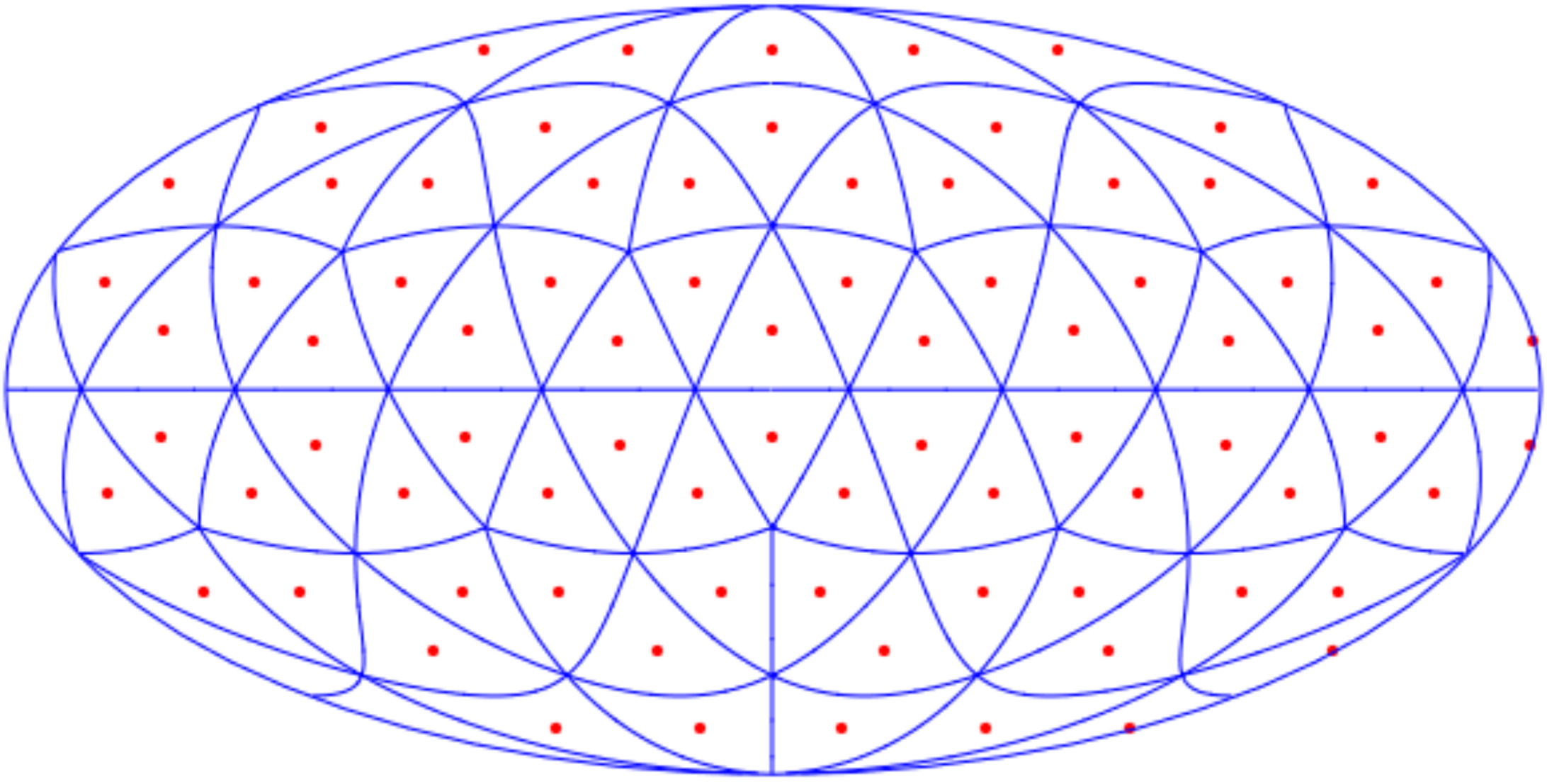
# Compressing the Gaia datasets

$2 \times 10^9$ STARS TIMES $10^2$ MEASUREMENTS = A LOT OF DATA

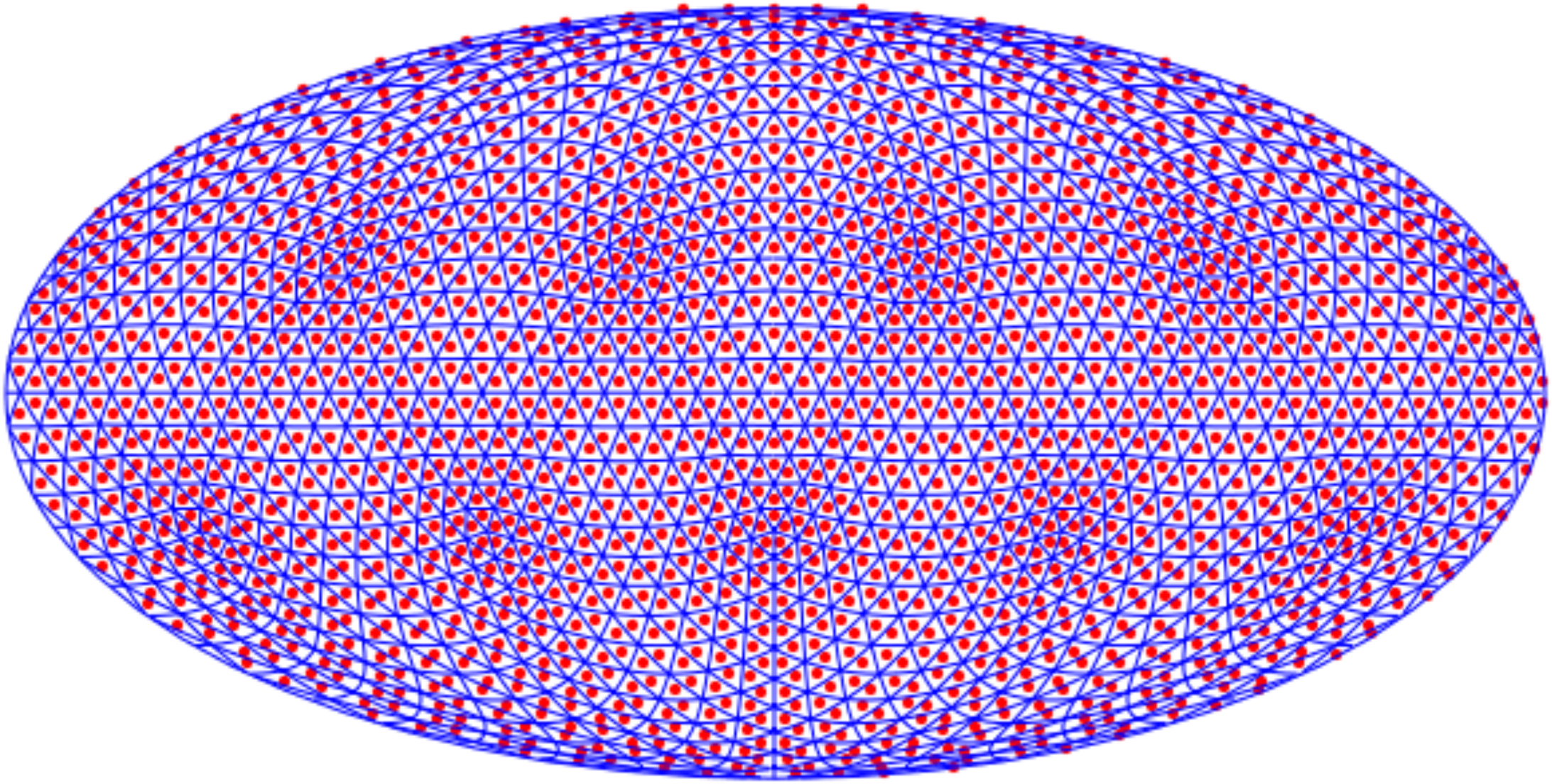
# Geodesic Dome



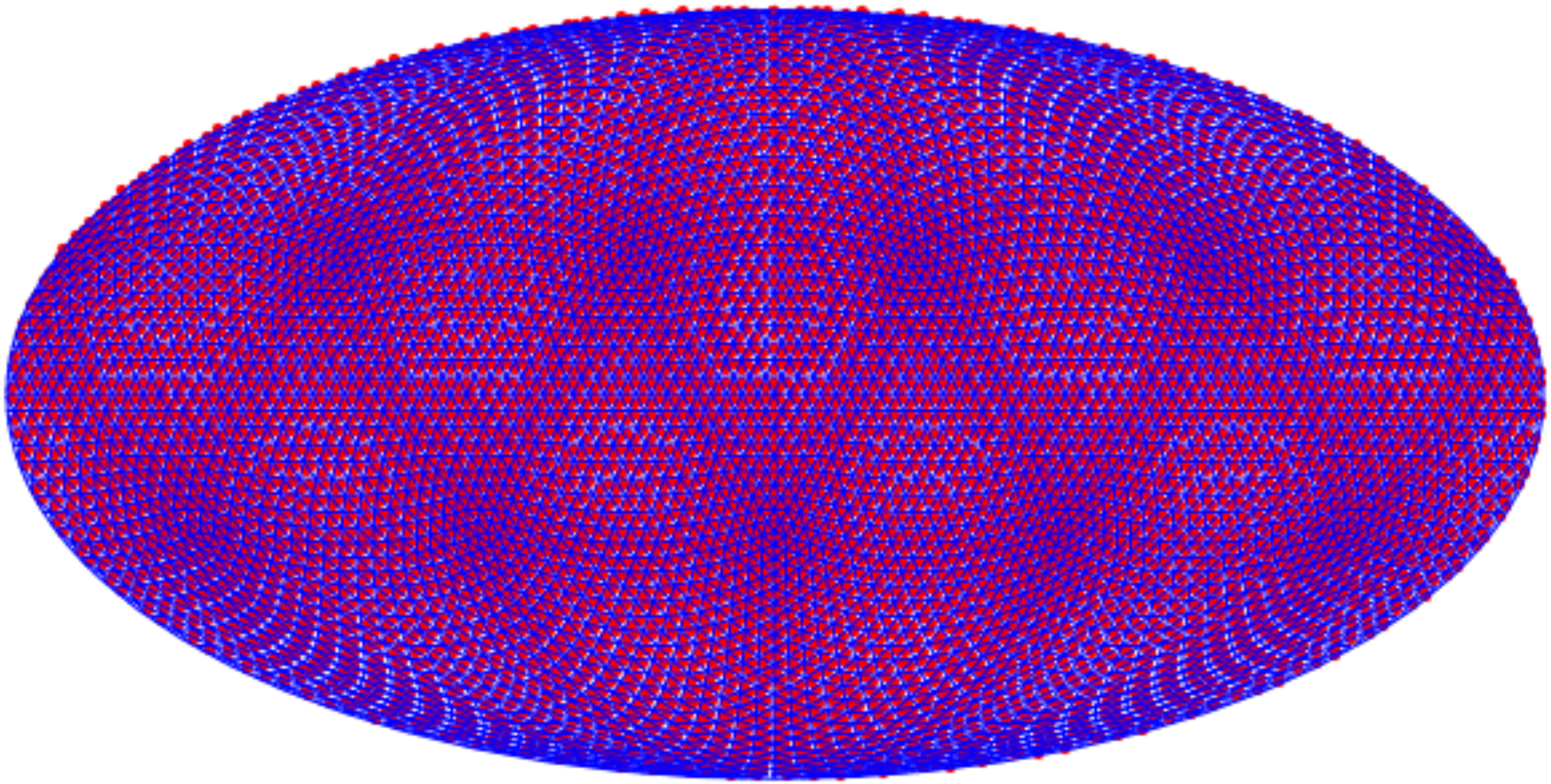




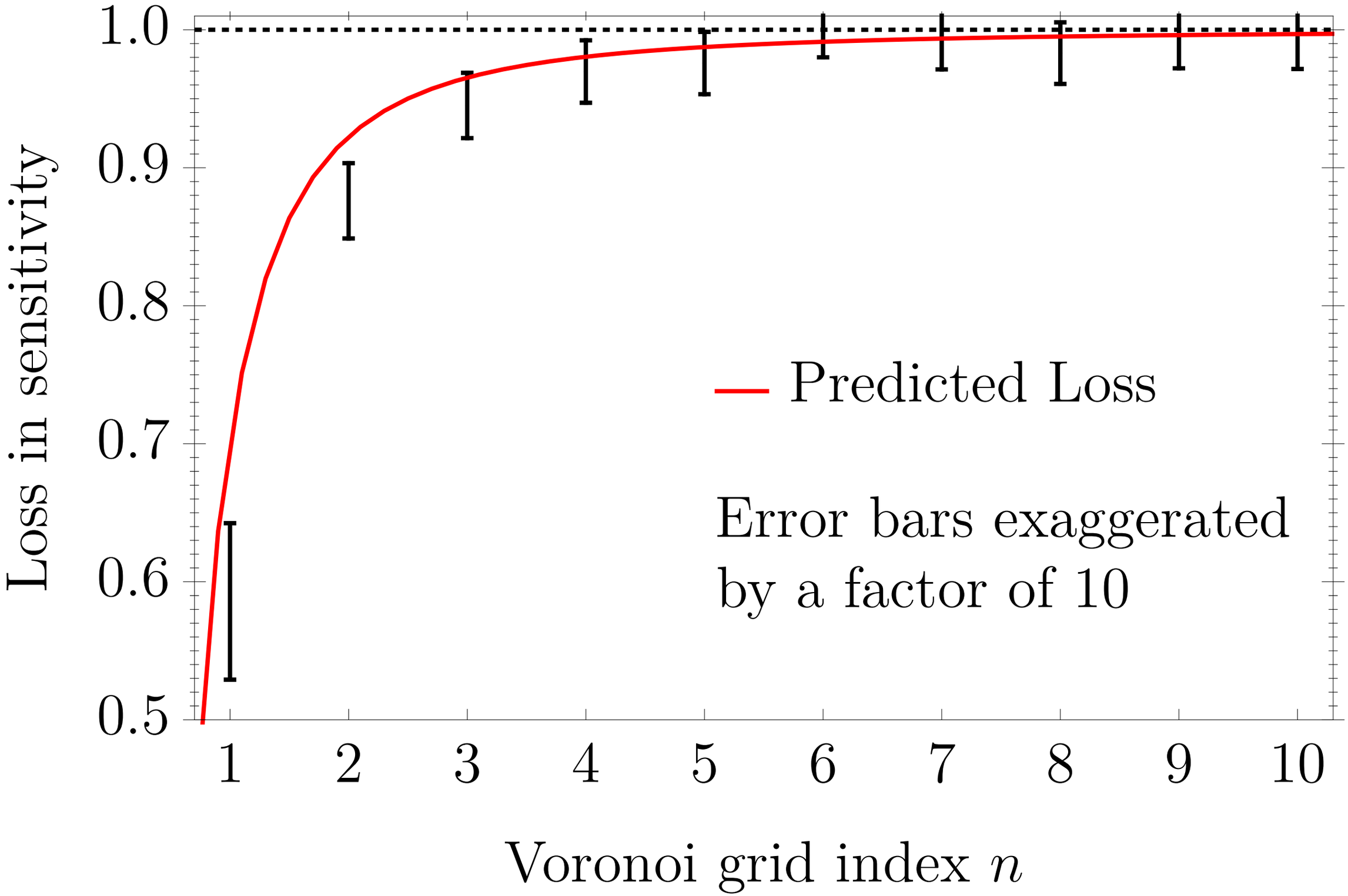
N = 2



N = 10



N = 20


1.0
0.9
0.8
0.7
0.6
0.5
Loss in sensitivity
1
2
3
4
5
6
7
8
9
10
Voronoi grid index $n$
Predicted Loss
Error bars exaggerated
by a factor of 10